

NX600 Series Communication Interface

(027-004-01E)

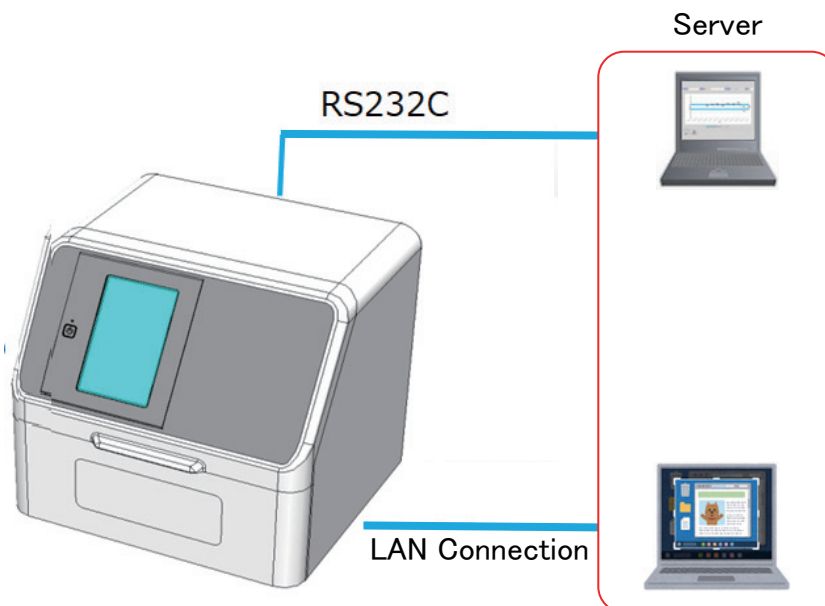
Revision History

No.	Issue Date	Reason	Pages Affected
1	Oct. 31, 2020	First release	All Pages

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1. System Configuration



2. Interface Specifications

2.1 Communication Interface

2.1.1 Outline

Communication interface has 7 types of as the following. In addition to Type 1 to Type 3 that have the same protocol as the FDC4000, Type 4 is added to support the HL7 standard, and Type A to Type C are added to support the Europe POCT.

The protocols except for Type 4 and Type C are able to use either a RS-232C connector (9pins) or a LAN connector alternatively.

Type 4 and Type C use only the LAN connector, and they do not support the serial connector.

(1) 2-way communication (requesting test information)

- Type 1 :

The analyzer requests test information (patient names, test names, etc.) to PC and sends test results after the measurements completed.

- Type A :

As a test result, test result of Type 1 including the following information is send. [Operator ID, Emergency test information, Slide lot, Slide valid term, Analyzer ID] Specification other than the above is the same as Type 1.

(2) One-way communication (without requesting test information) :

- Type 2 :

Type 2 is a one-way transmission of test results and its protocol is the same as Type 1.

- Type B :

As a test result, test result of Type 2 including the following information is send. [Operator ID, Emergency test information, Slide lot, Slide valid term, Analyzer ID] Specification other than the above is the same as Type 2 (Type 1).

(3) One-way communication (conventional) [Type 3]

Type 3 is one-way transmission as same as the protocol of the FDC4000.

(4) One-way communication conforming to HL7 standard (conventional) [Type 4 / Type C]

Type 4 / Type C is one-way transmission conforming to HL7 and is a protocol of LAB-29 of the IHE LAW profile. (Refer to DRI-CHEM NX600 Communication Specifications Appendix HL7 Specifications)

2.1.2 Serial

Com type	Type 1	Type 2	Type 3
Transmission path	RS-232C interface		
Synchronizing method	Asynchronous communication method		
Data transfer rate	19200 bps	19200 bps	9600 bps
Character length	8 bits		
Stop bit	1 bit		
Parity	None		
Flow control	Hardware method (RTS/CTS)		
BCC (Parity BCC)	BCC (Exclusive OR except for STX)		None

* Type 4 and Type C can not be used with Serial communication.

2.1.3 LAN

In the LAN connection, the analyzer is a client, and the PC is a server.

The IPv4 is supported, but the IPv6 is not supported.

The analyzer specifies the IP address and the port of the PC and saves the settings in the nonvolatile memory.

When the analyzer initializes the host connection by LAN, it perform the socket connection using the IP address and the port of the PC.

* For type 4 and Type C with LAN communication specification, please refer to "DRI-CHEM NX600 Communication Specifications Appendix HL7 Specifications".

2.2 Control Specifications

2.2.1 Time-out monitoring for transmission (Serial)

Condition: CTS signal remains inactive continuously.

Action: The analyzer clears the request and waits the next event.

2.2.2 Time-out monitoring for receiving reply (Serial)

Condition: Five seconds have passed without receiving the reply after sending the request command.

Action: The analyzer clears the request and waits the next event.

2.2.3 Re-transmission (Serial)

No re-transmission supported

3. Communication Data Format

3.1 Communication Data System

3.1.1 Text format

[STX (02H)], Command type, Parameter1, Parameter2, ... [ETX (03H)] [BCC]

The parameters are added as required.

The “ , ” (breakpoint) is inserted between a command type and a parameter and between parameters.

To connect some blocks, [ETB (17H)] is added at last.

Within parameter, “ , ” and “ @ ” are prohibited. Usable character codes are 20H to 7EH and A1H to DFH.

Ex.)

STX	Command type	,	Parameter 1	,	Parameter 2	ETX	BCC
-----	-----------------	---	----------------	---	----------------	-----	-----

Ex.)

STX	Command type	,	Parameter 1-1	,	Parameter 1-2	ETB		
			Parameter 2-1	,	Parameter 2-2	ETX	BCC	

3.1.2 BCC

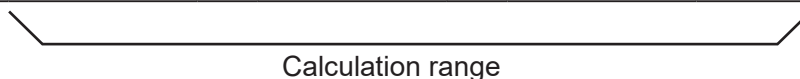
BCC is supposed to be 1 byte.

The calculation range for BCC is from the next of STX (command type) to ETX.

The calculation method is exclusive OR.

Ex.)

STX	Command type	,	Parameter 1	,	Parameter 2	ETX	BCC
-----	-----------------	---	----------------	---	----------------	-----	-----



For both the serial and LAN connections, BCC is provided for Type 1, Type 2, Type A and Type B, and it is not provided for Type 3.

3.2 Communication Data

3.2.1 Command type list

Request/Info (Analyzer=>PC)		Reply (PC=>Analyzer)	
Command type	Description	Command type	Description
W	Request for sample info	W	Reply for sample info
I	Request for Worklist index	I	Reply for Worklist index
S	Test start info		
R	Test results info		
E	Error info		

3.2.2 Definition

Name	Description
Command type	Given by one alphabetical character
Sample No. ^{*1*2}	Assigned sample No. given by PC
Patient ID ^{*2}	Unique ID for a patient (e.g patient's file No.)
Patient name	Patient name
Sample type	W or P or U
Number of test	Number of tests for a sample
Test name	Requested test name
Worklist index	A set of a sample No., a patient ID, a patient name and others
Species	Unique No. assigned for species
Reception No.	No. for a patient (unique No. within a working day). Unused
Operator ID	ID of the operator who performs the test
Test information that uses the emergency ID	Existence of the test that uses the emergency ID
Analyzer ID	Shipment machine number of the analyzer that performs the test
Slide lot	Lot code of the tested slide
Valid term	Valid term of the tested slide

*1 Sample No.: When PC assigns a number, the format is "yyyymmdd**" ["Year" + "Month" + "Day" + "Number (starting 01)"]. (Ex. 122001)

In the off-line mode, the format is "***" [Number (starting 01)] without year, month and day.

*2 : "Sample No." that is specified in this communication interface is displayed as "No." on the NX600 operation screen. In addition, "Patient ID" is displayed as "Sample ID". (Refer to 6 Correspondence Between Telegrams and the NX600 Operation Screen)

4. Communication Data Details

The support list of the communication data every type is shown below.

(O: Supported, -: Not supported)

Command type	Description	Type 1	Type 2	Type 3	Type A	Type B
W	Request for sample info	O	-	-	O	-
W	Reply for sample info	O	-	-	O	-
I	Request for Worklist index	O	-	-	O	-
I	Reply for Worklist index	O	-	-	O	-
S	Test start info	O	-	-	O	-
R	Test results info	O	O	O	O	O
E	Error info	O	-	-	O	-

A command send-timing is shown below.

The processing timing	The command use existence or non-existence				
	Request/Reply for sample info	Request/Reply for Worklist index	Test start info	Test results info	Error info
The work list acquisition	O (Type 1 / Type A)	O (Type 1 / Type A)	-	-	-
Normal measurement	-	-	O (Type 1 / Type A)	O	-
The retransmission from the past test record	-	-	-	O	-
The calibration	-	-	-	-	-
The control measurement	-	-	O (Type 1 / Type A)	O	-
The retransmission from the control measurement record	-	-	-	O	-
The error occurrence	-	-	-	-	O (Only for Type 1 and Type A, and the error that is outputted to the host)

4.1 Request for Worklist index

This is used when the analyzer requests index information of the Worklist from PC.

Item	Character	Size	Description	Parameter
Header	STX	1	STX: 02h	Required
Command type	'I'	1	Request command for Worklist index ('I': 49h)	Required
Breakpoint	','	1		Required
Sample No.	Numerical	13	Max. 13 characters	Optional (empty allowed)
Breakpoint	','	1		Required
Number of requests	Numerical	2	Maximum number of indexes to be requested (1 to 99)	Required
Delimiter	ETX	1	ETX: 03h	Required
BCC		1		Required

NOTE: Sample No. is the first key to search the Worklist. Blank data is allowable.

NOTE: The search direction is forward direction of the Worklist. But, the Worklist data which has already been informed to start the measurements will be searched last.

NOTE: The search results will be sent with the reply for the request (Worklist index).

<Ex.1> When requesting index information of 3 samples starting from sample No. "061201":

(Transmit data)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	STX	I	,	0	6	1	2	0	1	,	3	ETX	BCC		

□ = Space

<Ex.2> When requesting index information of 3 samples without assigning a starting sample No. for search:

(Transmit data)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	STX	I	,	,	3	ETX	BCC								

□ = Space

4.2 Reply for request (Worklist index)

This is used when PC sends index information of the Worklist to the analyzer.

Item	Character	Size	Description	Parameter
Header	STX	1	STX: 02h	Required
Command type	'I'	1	Request command for Worklist index ('I': 49h)	Required
Breakpoint	','	1		Required
Number of indexes	Numerical	2	Number of following indexes	Required (variable length)
Breakpoint	','	1		Required
Sample No. 1	Numerical	13	Max. 13 characters	Required (variable length)
Breakpoint	','	1		Required
Patient ID 1	Alphabetical & numerical	13	Max. 13 characters	Optional (variable length: empty allowed)
Breakpoint	','	1		Required
Patient name 1	Alphabetical & numerical	13	Max. 13 characters	Optional (variable length: empty allowed)
Breakpoint	','	1		Required
Species 1	Numerical	2	0 to 99 (Max. 2 characters)	Required (variable length)
Breakpoint	','	1		Required
Sex 1	Numerical	1	0: Male, 1: Female, 9: Undefined	Required
Breakpoint	','	1		Required
Age 1	Numerical	3	999: Undefined (Max. 3 characters)	Required (variable length)
Block breakpoint	ETB	1		Required for multiple samples
...				
Sample No. n	Numerical	13	Max. 13 characters	Required (variable length)
Breakpoint	','	1		Required
Patient ID n	Alphabetical & numerical	13	Max. 13 characters	Optional (variable length: empty allowed)
Breakpoint	','	1		Required
Patient name n	Alphabetical & numerical	13	Max. 13 characters	Optional (variable length: empty allowed)
Breakpoint	','	1		Required
Species n	Numerical	2	0 to 99	Required (variable length)
Breakpoint	','	1		Required

Item	Character	Size	Description	Parameter
Sex n	Numerical	1	0: Male, 1: Female, 9: Undefined	Required
Breakpoint	;	1		Required
Age n	Numerical	3	999: Undefined	Required (variable length)
Delimiter	ETX	1	ETX: 03h	Required
BCC		1		Required

NOTE: Number of indexes is essential.

NOTE: When no data in the Worklist, the "Number of indexes" should be 0 (zero). However, the sample No. send with the "Request for Worklist index" should be copied into the "Sample No. 1" field.

NOTE: When the Worklist has test orders, the sample No. is essential, and either the patient ID or the patient name is essential in the Worklist data.

NOTE: In case that the sample No. is blank, or both the patient ID and the patient name are blank (even if the sample No. has data), the field is invalid.

NOTE: If a number of index fields is larger than the "Number of indexes", overflowed fields (larger than the number of indexes) are ignored.

NOTE: If a number of index fields is smaller than the "Number of indexes", the process will continue without error.

NOTE: A block breakpoint (ETB) must NOT be added at the end of the last index field.

(Ex.) When replying with 2 samples data:

First sample information	Sample No.	2006061201
	Patient ID	ABCDEFGHJKLM
	Patient name	Taro Fuji
	Species	2: Feline
	Sex	1: Female
	Age	3 years
Second sample information	Sample No.	2006061202
	Patient ID	12345ABCD
	Patient name	Lucy Smith
	Species	1: Canine
	Sex	0: Male
	Age	1 year

(Transmit data)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	STX	I	,	2	,	2	0	0	6	0	6	1	2	0	1
2	,	A	B	C	D	E	F	G	H	I	J	K	L	M	,
3	T	a	r	o	□	F	u	j	i	,	2	,	1	,	3
4	ETB	2	0	0	6	0	6	1	2	0	2	,	1	2	3
5	4	5	A	B	C	D	,	L	u	c	y	□	S	m	i
6	t	h	,	1	,	0	,	1	ETX	BCC					

□ = Space

4.3 Request for sample information

This is used when the analyzer requests sample information of the Worklist from PC.

Item	Character	Size	Description	Parameter
Header	STX	1	STX: 02h	Required
Command type	'W'	1	Request command for sample info ('W': 57h)	Required
Breakpoint	','	1		Required
Sample No.	Numerical	13	Max. 13 characters	Required (variable length)
Delimiter	ETX	1	ETX: 03h	Required
BCC		1		Required

NOTE: The character data length for sample No. is max. 13 characters. If the data length is under 13, it is NOT necessary to fill the data area with "space".

(Ex.) When requesting sample information for the 2nd sample:

2nd sample information	Sample No.	2006061202
	Patient ID	12345ABCD
	Patient name	Tarou Fuji

(Transmit data)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	STX	W	,	2	0	0	6	0	6	1	2	0	2	ETX	BCC

□ = Space

4.4 Reply for request (sample info)

This is used when PC sends sample information of the Worklist to the analyzer.

Item	Character	Size	Description	Parameter
Header	STX	1	STX: 02h	Required
Command type	'W'	1	Request command for sample info ('W': 57h)	Required
Breakpoint	','	1		Required
Sample No.	Numerical	13	Max. 13 characters	Required (variable length)
Breakpoint	','	1		Required
Patient ID	Alphabetical & numerical	13	Max. 13 characters	Optional (variable length: empty allowed)
Breakpoint	','	1		Required
Patient name	Alphabetical & numerical	13	Max. 13 characters	Optional (variable length: empty allowed)
Breakpoint	','	1		Required
Number of tests	Numerical	2	0 to 30 tests (Max. 2 digits: '0' to '30') *This indicates the number of tests. Max. number is 30, which is a sum of 22 (max. number of tests by 1 sample) and 8 (max. number of calculated parameters).	Required (variable length)
Breakpoint 1	','	1		Required *Refer to the NOTE under this list
Test name 1	Alphabetical	8	Max. 8 characters	Required (variable length)
...				
Breakpoint n	','	1		Required
Test name n	Alphabetical	8	Max. 8 characters	Required (variable length)
Delimiter	ETX	1	ETX: 03h	Required
BCC		1		Required

Test name field

NOTE: Sample No. and number of tests are essential.

NOTE: If the "Number of tests" is "0", it means the Worklist does not have test orders.

NOTE: Maximum number of test name is 30.

NOTE: If a number of test name fields is larger than the "Number of tests", overflowed fields (larger than the number of tests) are ignored.

NOTE: If a number of test name fields is smaller than the "Number of tests", the process will continue without error.

NOTE: When the measurement number is 0, the assignment of “,” to breakpoint 1 makes it optional. Both “...,0,[ETX]” and “...,0[ETX]” are acceptable.

NOTE: <Processing of received data>

The received data (“Number of tests” and/or “Test name”) is used as printout data for the built-in printer, and then will be deleted after the printout completed.

To re-print out the information, perform from the beginning of the process (“Request for Worklist index”).

Also, the received data will not be used other purposes; e.g. for checking test results matching with the received data.

<Ex. 1> When replying for sample information of the 2nd sample:

2nd sample information	Sample No.	2006061202
	Patient ID	12345ABCD
	Patient name	Lucy Smith
	Number of tests	4 tests
	Test names	BUN, CRE, GLU, ALP

(Transmit data)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	STX	W	,	2	0	0	6	0	6	1	2	0	2	,	1
2	2	3	4	5	A	B	C	D	,	L	u	c	y	□	S
3	m	i	t	h	,	4	,	B	U	N	,	C	R	E	,
4	G	L	U	,	A	L	P	ETX	BCC						

□ = Space

1

<Ex. 2> When replying for sample information of the 1st sample:

1st sample information	Sample No.	2006061201
	Patient ID	ABCDEFGHIJKLM
	Patient name	Taro Fuji
	Number of tests	1
	Test names	Panel A

(Transmit data)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	STX	W	,	2	0	0	6	0	6	1	2	0	1	,	A
2	B	C	D	E	F	G	H	I	J	K	L	M	,	T	a
3	r	o	□	F	u	j	i	,	1	,	P	a	n	e	l
4	□	A	ETX	BCC											

□ = Space

4.5 Test start information

This is used when the analyzer sends test start information to PC.

Item	Character	Size	Description	Parameter
Header	STX	1	STX: 02h	Required
Command type	'S'	1	Test start information command('S': 53h)	Required
Breakpoint	,	1		Required
Test condition	Alphabetical	7	Selecting control measurements or normal measurements: CONTROL or NORMAL□ <□ : Space (20h)>	Required
Breakpoint	,	1		Required
Test date	Numerical	10	Year, month, and day when the measurements performed. (Ex. 2006-09-25)	Required
Breakpoint	,	1		Required
Test time	Numerical	5	Time when the measurements performed. (Ex. 10:50)	Required
Breakpoint	,	1		Required
Sample No.	Numerical	13	Unique ID for each sample	Required
Breakpoint	,	1		Required
Patient ID	Alphabetical & numerical	13	Unique ID for each patient	Required
Breakpoint	,	1		Required
Patient name	Alphabetical & numerical	13	Patient name	Required
Breakpoint	,	1		Required
Sample position	Numerical	2	Sample position on the sample disk	Required
Delimiter	ETX	1	ETX: 03h	Required
BCC		1		Required

NOTE: Each data field is a fixed-length field, so that data should be embedded from the left and spaces (20h) should be embedded in the remaining field.

<Ex.> When the test for the 1st sample starts on June 12, 2006 at 10:50:

Sample No.	2006061201
Patient ID	ABCDEFGHIJKLM
Patient name	Taro Fuji

(Transmit data)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	STX	S	,	N	O	R	M	A	L	□	,	2	0	0	6
2	-	0	6	-	1	2	,	1	0	:	5	0	,	2	0
3	0	6	0	6	1	2	0	1	□	□	□	,	A	B	C
4	D	E	F	G	H	I	J	K	L	M	,	T	a	r	o
5	□	F	u	j	i	□	□	□	□	,	0	1	ETX	BCC	

□ = Space

4.6 Test results information

4.6.1 Type 1 and Type 2

This is used when the analyzer sends test results to PC.

Item	Character	Size	Description	Parameter
Header	STX	1	STX: 02h	Required
Command type	'R'	1	Test results inform command ('R': 52h)	Required
Breakpoint	','	1		Required
Test condition	Alphabetical	7	Selecting control measurements or normal measurements (CONTROL or NORMAL□) <□ : Space (20h)>	Required
Breakpoint	','	1		Required
Test date	Numerical	10	Year, month, and day when the measurements performed. (Ex. 2006-09-25)	Required
Breakpoint	','	1		Required
Test time	Numerical	5	Time when the measurements performed. (Ex. 10:50)	Required
Breakpoint	','	1		Required
Sample No.	Numerical	13	Unique ID for each sample	Required
Breakpoint	','	1		Required
Patient ID	Alphabetical & numerical	13	Unique ID for each patient	Required
Breakpoint	','	1		Required
Patient name or Control solution type name	Alphabetical & numerical	13	When the test condition is NORMAL: Patient name When the test condition is CONTROL: Control solution type name	Required
Breakpoint	','	1		Required
Species	Numerical	2	0 to 99 [2 characters fixed (padding by 0)]	Required
Breakpoint	','	1		Required
Sex	Numerical	1	0: Male, 1: Female, 9: Stand-alone mode	Required
Breakpoint	','	1		Required
Age	Numerical	3	999: Stand-alone mode 3 characters fixed (padding by 0)	Required
Breakpoint	','	1		Required
Sample position	Numerical	2	Sample position on the sample disk. * 01 fixed (NX600)	Required
Breakpoint	','	1		Required
Number of tests	Numerical	2	0 to 30 (2 characters fixed (padding by 0)) *This indicates the number of tests. Max. number is 30, which is a sum of 22 (max. number of tests by 1 sample) and 8 (max. number of calculated parameters).	Required
Breakpoint	','	1		Required

Item	Character	Size	Description	Parameter
Test name	Alphabetical & numerical	8	Test name (5 characters) + ' - ' + Sample type (2 characters) Whole blood: -W, Plasma/Serum: -P, Urine: -U.	Required
Breakpoint	','	1		Required
Equal sign		1	Equal sign or unequal signs	Required
Breakpoint	','	1		Required
Test result	Alphabetical & numerical	9	Test result (9 characters)	Required
Unit	Alphabetical & numerical	6	Unit (6 characters)	Required
Breakpoint	','	1		Required
Dilution factor	Numerical	2	Used dilution factor (from 01) * The calculated parameter is expressed as 01 time of numeral.	Required
Breakpoint	','	1		Required
Reference interval lower limit	Numerical	5	Setting of reference interval lower limit	Required
Breakpoint	','	1		Required
Reference interval upper limit	Numerical	5	Setting of reference interval upper limit	Required
Breakpoint	','	1		Required
Warning	Alphabetical	11	Test warning. Refer to 4.6.5 Warning characters.	Required
Test name breakpoint	','	1	Breakpoint	Required
Delimiter	ETX	1	ETX: 03h	Required
BCC		1		Required

Test result field

NOTE: The test result field has "Number of tests" of test results.

NOTE: A test name breakpoint must NOT be added to the test result field of the last test slide.

NOTE: Each data area is a fixed-length field, so that data should be embedded from the left and spaces (20h) should be embedded in the remaining spaces.

NOTE: "Sex" and "Age" fields

In the 2-way communication, the data send with the "Request for Worklist index" should be copied into the "Sex" and "Age" fields.

In the stand-alone mode, the "Sex" field should be embedded with "9", and the "Age" field with "999".

Transmission example

<Transmit data>

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	STX	R	,	N	O	R	M	A	L	□	,	2	0	0	6
2	-	0	6	-	1	2	,	1	0	:	5	0	,	2	0
3	0	6	0	6	1	2	0	1	□	□	□	,	A	B	C
4	D	E	F	G	H	I	J	K	L	M	,	T	a	r	o
5	□	F	u	j	i	□	□	□	□	,	0	2	,	1	,
6	0	0	3	,	0	1	,	0	2	,	G	L	U	-	P
7	S	□	□	,	=	,	7	5	□	□	□	□	□	□	□
8	m	g	/	d	l	□	,	1	0	,	5	0	.	0	□
9	,	1	0	0	.	0	,	□	@	#	+	*	□	□	□
10	E	□	□	,	A	M	Y	L	-	P	S	□	,	>	,
11	1	5	0	0	□	□	□	□	□	u	/	L	□	□	□
12	,	0	1	,	5	0	0	□	□	,	1	5	0	0	□
13	,	H	□	#	□	□	□	□	□	□	□	□	ETX	BCC	

□ = Space

4.6.2 Type 3

This is used when the analyzer sends test results to PC.

Item	Character	Size	Description	Parameter
Header	STX	1	STX: 02h	Required
Test condition	Alphabetical	7	Selecting control measurements or normal measurements CONTROL or NORMAL□ <□ : Space (20h)>	Required
Test date	Numerical	10	Year, month, and day when the measurements performed. (Ex. 2006-09-25)	Required
Test time	Numerical	5	Time when the measurements performed. (Ex. 10:50)	Required
Sample No.	Numerical	13	Unique ID for each sample	Required
Patient ID	Alphabetical & numerical	13	Unique ID for each patient	Required
Sample position	Numerical	2	Sample position on the sample disk. * 01 fixed (NX600)	Required

--- Go to the next page---

--- Continued from the previous page.---

Item	Character	Size	Description	Parameter
Test name/ Sample type	Alphabetical & numerical	7	Test name (5 characters) + ' - ' + Sample type (2 characters) Left-justified (The space is embedded in the right remaining field.) Whole blood: -W, Plasma/Serum: -P, Urine: -U. "Blank: -E" is not exist.	Required
Equal sign		1	Equal sign (=) or unequal signs (< >)	Required
Test result	Alphabetical & numerical	9	Test result (9 characters) Left-justified (The space is embedded in the right remaining field.)	Required
Unit	Alphabetical & numerical	6	Unit (6 characters) Left-justified (The space is embedded in the right remaining field.)	Required
Dilution factor	Numerical	2	Used dilution factor (from 01) *The calculated parameter is expressed as 01 time of numeral.	Required
Warning	Alphabetical	11	Test warning. Refer to 4.6.5 Warning characters	Required
Delimiter	ETX	1	ETX: 03h	Required

Test result field

NOTE: The test result field has "Number of tests" of test results.

<Transmit data>

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
STX	N	O	R	M	A	L	□	2	0	1	4	-	0	7	-	2	5	1	7
:	5	9	1	2	3	4	5	6	7	8	9	0	1	2	3	A	B	C	D
E	F	G	H	I	J	K	L	M	0	1	G	L	U	-	P	□	□	=	1
2	3	4	5	□	□	□	□	m	g	/	d	l	□	1	5	□	@	#	+
*	F	□	□	E	\	□	A	M	Y	L	-	U	□	>	1	0	0	0	□
□	□	□	□	m	g	/	d	l	□	0	2	H	□	#	□	□	□	□	□
□	□	□	ETX																

□ = Space

Regardless of the serial or LAN, no BCC is given for Type 3.

4.6.3 type A / B

This is used when the analyzer sends test results to PC.

Item	Character	Size	Description	Parameter
Header	STX	1	STX: 02h	Required
Command type	'R'	1	Test results inform command ('R': 52h)	Required
Breakpoint	','	1		Required
Test condition	Alphabetical	7	Selecting control measurements or normal measurements (CONTROL or NORMAL□) <□ : Space (20h)> * QC is only for the NX500, so it is not used for the NX600.	Required
Breakpoint	','	1		Required
Test date	Numerical	10	Year, month, and day when the measurements performed. (Ex. 2006-09-25)	Required
Breakpoint	','	1		Required
Test time	Numerical	5	Time when the measurements performed. (Ex. 10:50)	Required
Breakpoint	','	1		Required
Sample No.	Numerical	13	Unique ID for each sample	Required
Breakpoint	','	1		Required
Patient ID	Alphabetical & numerical	13	Unique ID for each patient	Required
Breakpoint	','	1		Required
Patient name or Control solution type name	Alphabetical & numerical	13	It depends on the test condition. - When the test condition is NORMAL: Patient name When the test condition is CONTROL or QC:Control solution type name * Only the patient name is sent for the test start information. * QC is only for the NX500, so it is not used for the NX600.	Required
Breakpoint	','	1		Required
Species	Numerical	2	0 to 99	Required
Breakpoint	','	1		Required
Sex	Numerical	1	0: Male, 1: Female, 9: Stand-alone mode	Required
Breakpoint	','	1		Required
Age	Numerical	3	999: Stand-alone mode 3 characters fixed (padding by 0)	Required
Breakpoint	','	1		Required
Sample position	Numerical	2	Sample position on the sample disk. * 01 fixed (NX600)	Required
Breakpoint	','	1		Required

Item	Character	Size	Description	Parameter
Number of tests	Numerical	2	0 to 30 (2 characters fixed (padding by 0) *This indicates the number of tests. Max. number is 30, which is a sum of 22 (max. number of tests by 1 sample) and 8 (max. number of calculated parameters).	Required
Breakpoint	‘,’	1		Required
Operator ID	Alphabetical & numerical	13	Operator ID (13 characters)	Required
Breakpoint	‘,’	1		Required
Test information that uses the emergency ID	Numerical	1	0: No emergency ID is used 1: Emergency ID is used	Required
Breakpoint	‘,’	1		Required
Analyzer ID	Alphabetical & numerical	8	Shipment machine number (8 characters)	Required
Breakpoint	‘,’	1		Required
Test name	Alphabetical & numerical	8	Test name (5 characters) + ' - ' + Sample type (2 characters) Whole blood: -W, Plasma/Serum: -P, Urine: -U.	Required
Breakpoint	‘,’	1		Required
Equal sign		1	Equal sign or unequal signs	Required
Breakpoint	‘,’	1		Required
Test result	Alphabetical & numerical	9	Test result (9 characters) * When the emergency test is not performed, and the valid term of the slide has expired: EXP * When the slide is set to invalid: CHK QC The priority is EXP > CHK QC.	Required
Unit	Alphabetical & numerical	6	Unit (6 characters)	Required
Breakpoint	‘,’	1		Required
Dilution factor	Numerical	2	Used dilution factor (from 01) * The calculated parameter is expressed as 01 time of numeral.	Required
Breakpoint	‘,’	1		Required
Reference interval lower limit	Numerical	5	Setting of reference interval lower limit	Required
Breakpoint	‘,’	1		Required
Reference interval upper limit	Numerical	5	Setting of reference interval upper limit	Required
Breakpoint	‘,’	1		Required
Warning	Alphabetical	11	Test warning. Refer to 4.6.5 Warning characters	Required

Test result field

Item	Character	Size	Description	Parameter
Breakpoint	','	1		Required
Slide lot	Alphabetical & numerical	6	Slide lot code (6 characters) * There is no lot code for the calculated item, so fill it with spaces.	Required
Breakpoint	','	1		Required
Valid term	Numerical	6	Slide valid term YYYYMM (Example: 200609) * There is no valid term for the calculated item, so fill it with spaces.	Required
Test name breakpoint	','	1	Breakpoint	Required
Delimiter	ETX	1	ETX: 03h	Required
BCC		1		Required

Test result field

NOTE: The test result field has “Number of tests” of test results.

NOTE: A test name breakpoint must NOT be added to the test result field of the last test slide.

NOTE: Each data area is a fixed-length field, so that data should be embedded from the left and spaces (20h) should be embedded in the remaining spaces.

NOTE: “Sex” and “Age” fields

In the 2-way communication, the data send with the “Request for Worklist index” should be copied into the “Sex” and “Age” fields.

In the stand-alone mode, the “Sex” field should be embedded with “9”, and the “Age” field with “999”.

NOTE: The QC output for each test condition is performed in the following conditions during the control test.

Control test setting/USM	Yes	No
With correlation correction	QC□□□□□	CONTROL
Without correlation correction	CONTROL	CONTROL

1

4.6.4 Test result

The setting value of result of test is shown below.

Set value	Condition
Testing value	When test has been completed normally. (Except below)
Upper limit of measurement range	Colorimetric analysis, Immunological test : When the measurement value is higher than the upper limit of measurement range.
Lower limit of measurement range	Colorimetric analysis, Immunological test : When the measurement value is lower than the upper limit of measurement range.
+OR	ISE : When the measurement value is higher than the upper limit of measurement range.
-OR	ISE : When the measurement value is lower than the lower limit of measurement range.
****	When error has occurred.

4.6.5 Warning characters

The below table indicates warning character allocation in the warning data field.

When no error has occurred for each, a space (20h) will be embedded.

Position (sending order)	Warning character	Description
1st	H	Testing value exceeds the upper limit of the preset reference interval.
	L	Testing value falls below the lower limit of the preset reference interval.
2nd	@	Testing value is outside of the determination range.
3rd	#	The valid term of the slide has expired.
4th	\$	Temperature control error (thermistor disconnection)
	+	Temperature control error (Incubator temperature has exceeded the upper limit of the range.)
	-	Temperature control error (Incubator temperature has fallen below the lower limit of the range.)
5th	*	Photometer malfunction (white plate fluctuation)
	?	Photometer malfunction (out of the white plate range)
6th	Unused	Always a space (20h)
7th	&	Abnormally high testing value
8th	Unused	Always a space (20h)
9th	E	Testing value is error. (Calculation error or malfunction of ISE test results*)
10th	¥	Un-spotted slide
11th	Unused	Always a space (20h)

*< When the electrolyte measurement malfunction (Drift-Impedance-overrange error) occurs >
“E” is added to “9th” warning position.

4.6.6 Control solution type name characters

In the control solution type name, the following symbols can be used.

-#/?!.;()<>

(, comma) is not used because it cannot be entered on the analyzer.

4.7 Error information

This is used when the analyzer sends error information to PC.

Item	Character	Size	Description	Parameter
Header	STX	1	STX: 02h	Required
Command type	'E'	1	Error information command	Required
Breakpoint	','	1		Required
Error event date	Numerical	10	Year, month, and day when the error occurred. (Ex: 2006-06-12)	Required
Breakpoint	','	1		Required
Error event time	Numerical	8	Time when the measurements performed. (Ex: 10:30:50)	Required
Breakpoint	','	1		Required
Error number	Alphabetical & numerical	5	Error number	Required
Breakpoint	','	1		Required
Number of added info	Numerical	1	Number of added info (0 to 9)	Required
Breakpoint	','	1		Required
Added info 1	Alphabetical & numerical	6	Added info on the error (voltage, etc.)	Required
Breakpoint	','	1		Required
...				Required
Added info n	Alphabetical & numerical	6	Added info on the error (voltage, etc.)	Required
Delimiter	ETX	1	ETX: 03h	Required
BCC		1		Required

Added info field

1

NOTE: The added info field has "Number of added info" of data.

NOTE: A breakpoint must NOT be added at the end of the last added info field.

NOTE: Each data field is a fixed-length field, so that data should be embedded from the left and spaces (20h) should be embedded in the remaining field.

<Ex.> When suction clogging error (Error number: E0110/Number of added info: 1/Added info: 1.000) has occurred:

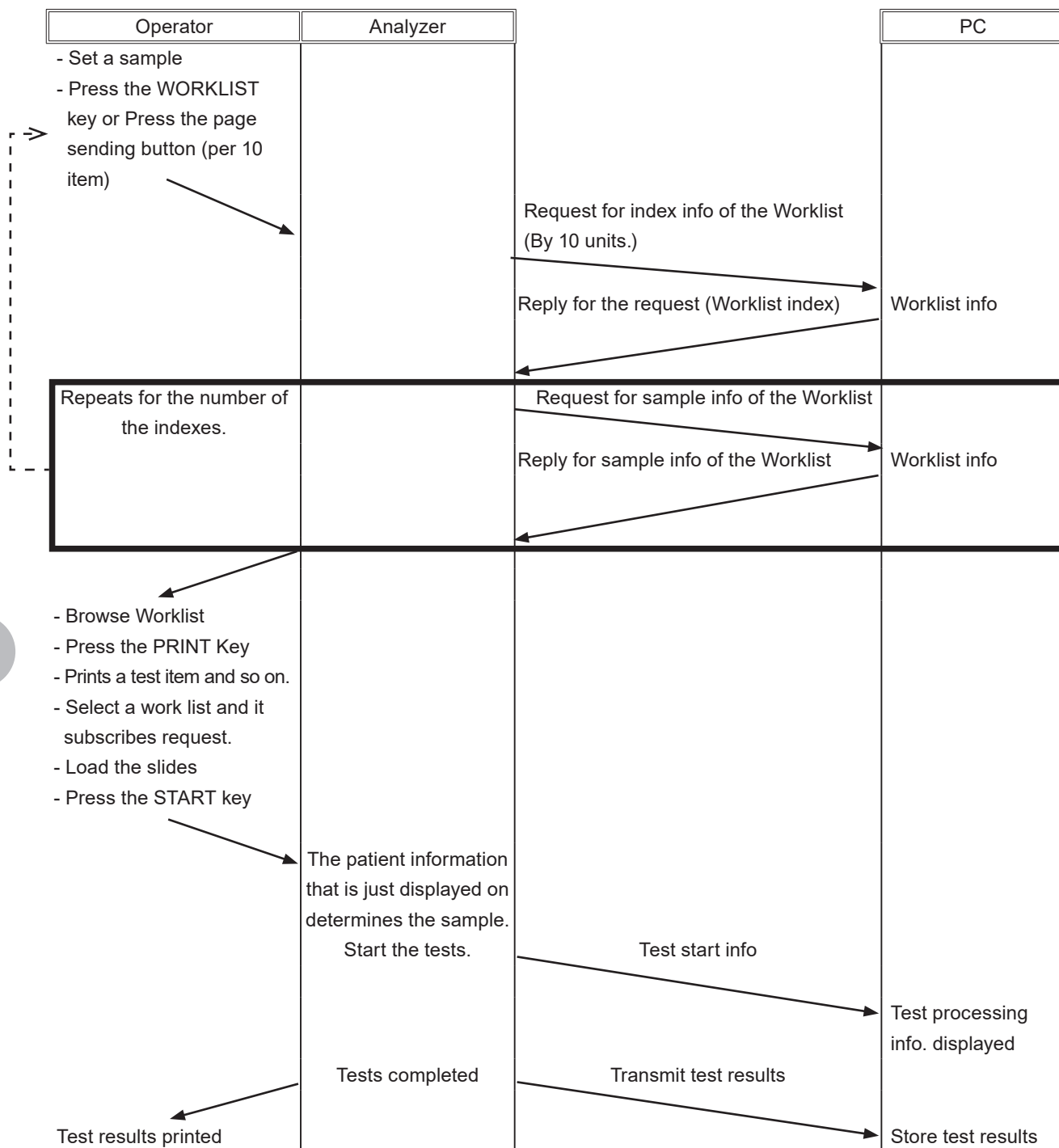
(Transmit data)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	STX	E	,	2	0	0	6	-	0	6	-	1	2	,	1
2	0	:	3	0	:	5	0	,	E	0	1	1	0	,	1
3	,	1	.	0	0	0	□	ETX	BCC						

□ = Space

5. Data Transmission Flow

5.1 Single sample test (Type 1)



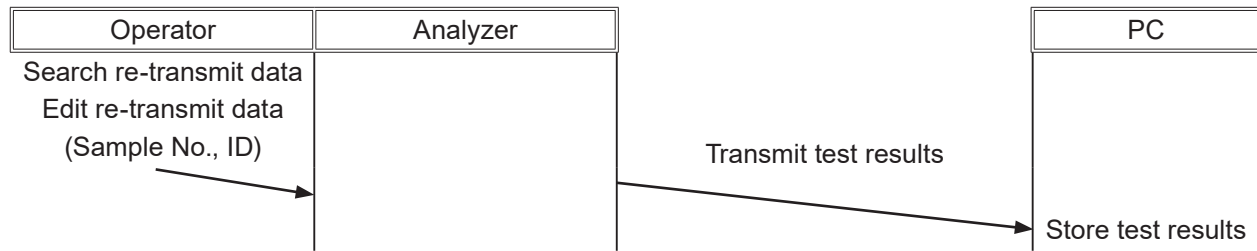
NOTE: Pressing the WORKLIST key requests Worklist index information and Detail information at the same time. (the 10 first samples)

NOTE: By the page sending it button push every 10 matters, it does the question and the detailed question of the index all together about the following 10 samples.

NOTE: By pushing a start-key, it begins the measurement of the maintained sample.

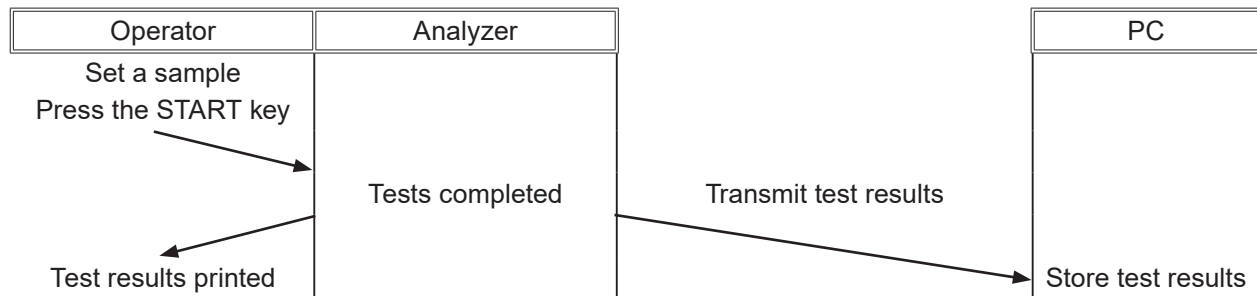
5.2 Re-transmitting test results (Type 1, Type 2, Type3)

This function is used to re-transmit test results manually. (This is useful when electric power failure or cable disconnection caused the communication interruption.)



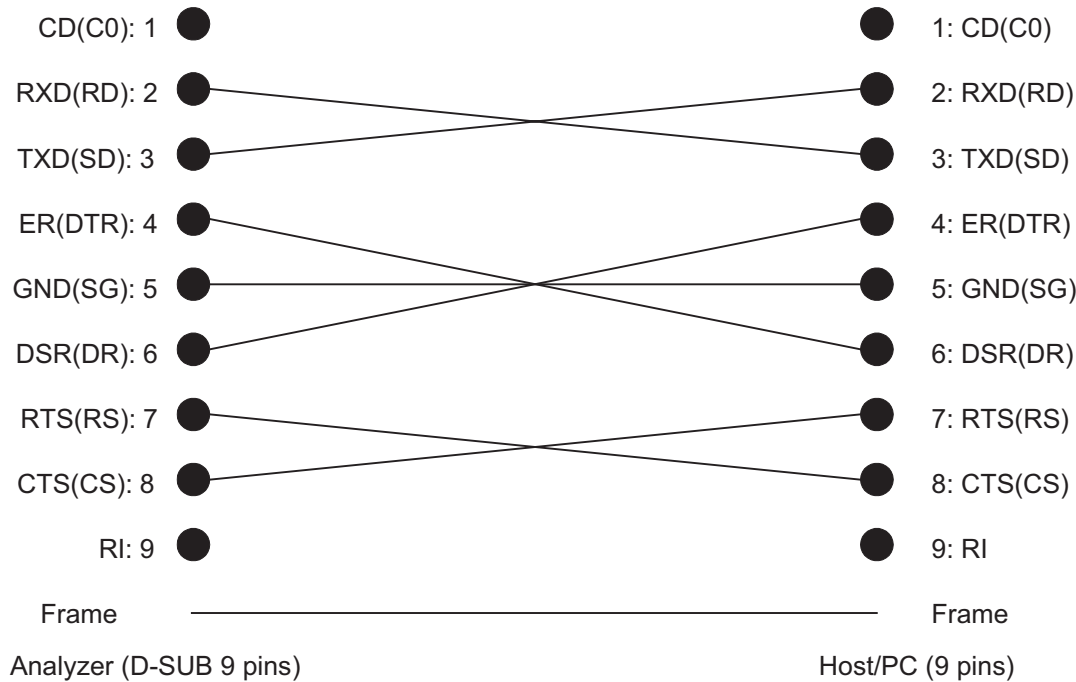
5.3 Transmitting test results (1-way communication:)

This is 1-way transmission of test results.



5.3.1 Communication Cable Connection

- (1) RS-232C cable
D-SUB 9 pins



- (2) LAN cable
Straight

1

6. Correspondence Between Telegrams and the NX600 Operation Screen

“Sample No.” that is specified in this communication interface is displayed as “No.” on the NX600 operation screen. In addition, “Patient ID” is displayed as “Sample ID”.

Telegrams	Screen display
Sample No.	No.
Patient ID	Sample ID

(Appendix) Test parameter name in electric message

	Parameter	Parameter name in electric message	
		Whole blood sample	Serum/plasma sample
Enzyme	ALP		ALP-P
	ALPi		ALPi-P or ALP-P
	AMYL		AMYL-P
	v-AMY		AMYL-P
	CHE		CHE-P
	CKMB		CKMB-P
	CPK		CPK-P
	GGT		GGT-P
	GOT/AST		GOT-P or AST-P
	GPT/ALT		GPT-P or ALT-P
	LAP		LAP-P
	LDH		LDH-P
	LDHi		LDHi or LDH-P
	LIP		LIP-P
	v-LIP		vLIP
General Chemistry	ALB		ALB-P
	BUN		BUN-P
	UREA		UREA-P
	Ca		Ca-P
	CRE		CRE-P
	DBIL		DBIL-P
	GLU		GLU-P
	HDL-C		HDL-C-P
	IP		IP-P
	Mg		Mg-P
	NH ₃	NH3-W	NH3-P
	TBIL		TBIL-P
	TCHO		TCHO-P
	TCO ₂		TCO2-P
	TG		TG-P
	TP		TP-P
	UA		UA-P
Electrolytes	Na	Na-W	Na-P
	K	K-W	K-P
	Cl	Cl-W	Cl-P
Immunological test	CRP		CRP-P
	vc-CRP		cCRP-P
Calculated parameter	LDL-C		LDL-C
	non-HDL-C		non-HDL
	GOT/GPT or AST/ALT		GOT/GPT or AST/ALT
	GLOB		GLOB
	ALB/GLOB		ALB/GLB
	Na/K		Na/K
	BUN/CRE		BUN/CRE
	UREA/CRE		UREA/CRE, UREA/Cr*
	Anion Gap		AGAP

* for Type3



DRI-CHEM NX600 Communication Specifications



Appendix HL7 Specifications

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1. Introduction

1.1 Purpose of the Document

FDC shall be compatible with the HL7 v2.5.1 message protocol and IHE Pathology and Laboratory Medicine Technical Framework^{*1} in order to exchange information with medical devices that conform to HL7 standards such as LAS and LIS on the network.

This document describes the HL7 standards compatible with FDC in the appendix of the FDC communication standards.

*1. Clinical guidelines that indicate how IHE uses the standard specifications prescribed according to the business scenario

1.2 Glossary

Definitions, terms and abbreviations used in this document are as follows.

Term	Description
HL7	There are instances where HL7 International Inc. and standard specifications of medical information exchange by HL7 are indicated. It is used for the latter in this document. Official website: http://www.hl7.org
LOINC	An item name and its code standard for information obtained through testing, consultation, etc., developed and managed by the Regenstrief Institute. Official website: https://loinc.org/
JLAC10	Clinical test item classification code established by the Japanese Society of Laboratory Medicine. JLAC10 code is 17 digits: 5 digits for analyte, 4 digits for identification, 3 digits for materials, 3 digits for measurement method and 2 digits for results identification. Official website: https://www.jslm.org/committees/code/#set4
OUL	Unsolicited laboratory observation message An unrequested measurement results message that is one kind of HL7 message exchanged between systems.
UCUM	the Unified code for units of measure The unit code set including all units defined by ISO 1000, ISO 2955-1983, ANSI X3.50-1986, HL7 and ENV 12435 Official website: http://unitsofmeasure.org/
IHE	Integrating the Healthcare Enterprise An international project to promote interconnectivity between medical information systems. Create guidelines that indicate how to use HL7 and other standard specifications. http://www.ihe.net/
JAHIS	Japanese Association of Healthcare Information Systems Industry Japanese association that implements improvement, ensures quality and safety, and promotes standardization for technology related to health and welfare information systems. Establishes data exchange protocol using HL7 as a JAHIS standard. Official website: https://www.jahis.jp/
Analyzer	FDC equipment.
Analyzer Manager	Equipment that manages orders, results, etc., of LAS, LIS and other FDC equipment.

1.3 HL7 interpretation: HL7 standards interpretation

Interprets standards based on the IHE guidelines^{*1}, JAHIS standards^{*2} and other standards that indicate HL7 protocol and use in conforming to HL7 protocol.

Additionally, segments and elements not deemed compulsory by HL7 protocol are set if there is settable information. If inconsistency arises in protocol interpretation, the FDC shall give priority to the IHC due to high overseas demand.

*1. Guidelines that indicate how to use standard specifications prescribed by IHE. This document refers to the IHE Pathology and Laboratory Medicine Technical Framework.

*2. Japanese standards prescribed by JAHIS. This document refers to the JAHIS Clinical Test Data Exchange Protocol.

<References>

- HL7 standard version 2.5.1
- IHE Pathology and Laboratory Medicine Technical Framework Revision 8.0 (June 2017)
- JAHIS Clinical Test Data Exchange Protocol Ver. 4.0C (July 2016)

1.4 HL7 version: HL7 version

HL7 is currently still being updated, and v2 and v3 series exist.

The most popular is the v2 series, with the most widely adopted being v2.5 in Japan and v2.5.1 overseas. FDC has selected v2.5.1 due to high overseas demand.

1.5 IHE Profile Support: IHE Integrated Profile Support List

Although the IHE has integrated profiles for every medical field, FDC supports Pathology and Laboratory Medicine Technical Framework for the clinical testing field. There are multiple integrated profiles within the technical framework, but FDC supports the following profiles related to IVD equipment.

- Laboratory Analytical Workflow (LAW)

1.5.1 Laboratory Analytical Workflow(LAW)

The LAW profile prescribes information exchange related to measurement requests and results between Analyzer (IVD equipment) and Analyzer Manager (LAS, LIS, etc.) equipment. There are three transactions in LAW, but FDC shall support LAB-29 (test results notification) only.

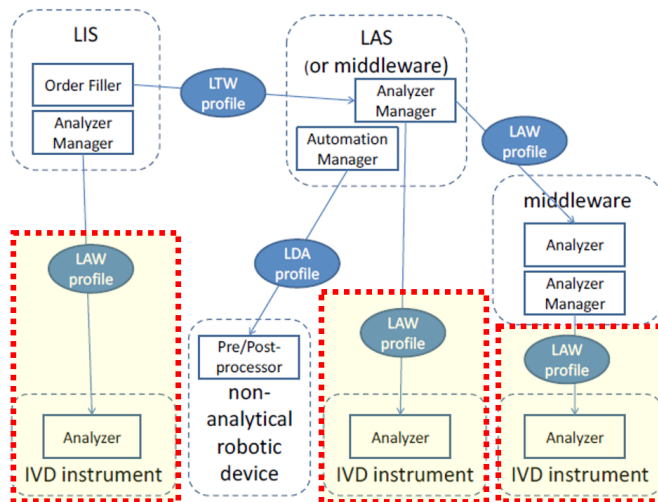


Figure X.3-1: Interconnection of systems and mapping to profile's actors

Relationship Between Equipment and the LAW Profile

X.4 Actors/ Transactions

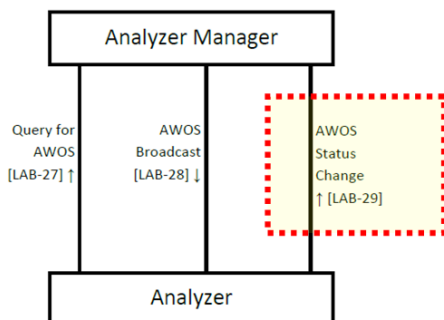


Figure X.4-1: Laboratory Analytical Workflow Actor Diagram

Supports LAB-29 (test results notification) only

1280 Table X.4-1 lists the transactions for each actor involved in the LAW Profile. To claim support of this Integration Profile, an implementation of an actor must perform the required transactions (labeled "R"). Transactions labeled "O" are optional and define the profile options explained in Section X.5 below.

1285 Table X.4-1: LAW Integration Profile - Actors and Transactions

Actors	Transactions	Optionality	Section in Vol. 2
Analyzer Manager	LAB-27 : Query for AWOS	R ¹	Q
	LAB-28 : AWOS Broadcast	R ³	R
	LAB-29 : AWOS Status Change	R ³	Y
Analyzer	LAB-27 : Query for AWOS	O ²	Q
	LAB-28 : AWOS Broadcast	O ^{2,3}	R
	LAB-29 : AWOS Status Change	R	Y

¹ An Analyzer Manager must support LAB-27(query) as the default mechanism for AWOS transfer.

² An Analyzer must support both LAB-27 and LAB-28 if supporting the bi-directional option. See X.5 below.

³ The transaction may contain enhanced information. The usage of the enhanced information is guided by the discussion in X.5 LAW Integration Profile Options and W.1.3 Profile Types.

Used Transaction

2. System Layout



3. Interface Specification

3.1 Communication Method

HL7 output shall have the following communication method and use LAN.

(1) Single direction communication of measurement results (type 4)

Outputs measurement results upon completion of measurement and when resending results.

3.1.1 LAN

Only IPv4 shall be supported, IPv6 is not supported.

The Analyzer Manager IP address and port shall be settable from the FDC (see 3.1.1.1), and setting values shall be stored.

3.1.1.1 IP Address and Port Number

The IP address and port number shall be settable in the mode function.

Item	Setting Location	Default Value
Analyzer	Host Connection Setting > FDC Setting	IP address: 192.168.100.1 Port number: 65535
Analyzer Manager	Host Connection Setting > PC Setting	IP address: 192.168.100.2 Port number: 65535

3.2 Lower Layer Network Protocol

Supports HL7 Minimal Lower Layer Protocol in accordance with HL7 specifications.

3.2.1 Setting MLLP

MLLP (Minimal Lower Layer Protocol) defines the header and footer of a message to clarify start and end positions of a message even in unstable connection environments such as a serial connection. It is not officially established as an HL7 standard, but shall be supported as it has been used for a long time between HL7 connected equipment.

MLLP shall attach SB (Start Block), EB (End Block) and CR (Carriage Return) to the front and back of messages.

<SB>	Message	<EB>	<CR>
------	---------	------	------

Element	Description
<SB>	ASCII <VT>, or, <0x0B>. Not to be confused with SOH or STX ASCII characters, etc.
Message	This shall be block HL7 data content. This data can contain any displayable ASCII character and carriage return character (<CR>).
<EB>	ASCII <FS>, or, <0x1C>. Not to be confused with ETX or EOT ASCII characters, etc.
<CR>	ASCII carriage return character, or, <0x0D>.

Shall be custom and full-compatible because the attachment method varies depending on the connection target. Each method shall be set with the Type 4 setting MLLP method.

MLLP Method	Format	Notes
Custom	Message <EB> <CR>	For Japanese domestic use. IHE-J is recommended.
Full	<SB> Message <EB> <CR>	For overseas use.

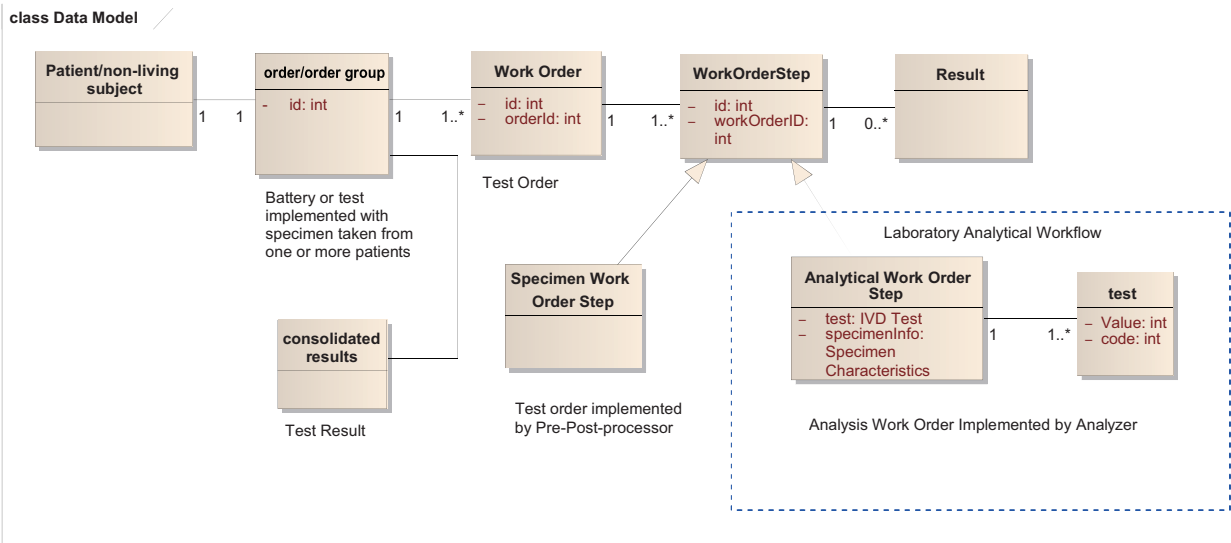
The received message shall also be interpreted by the set MLLP method.

3.3 Spec List

Item	Details	Reason	Setting Method
HL7			
Test Code Support	▪ The following shall be supported as test code used in HL7 messages - LOINC - JLAC10	Shall support both codes because Japan uses JLAC10 and overseas uses LOINC	Specification Reference for Each Product
Character Set Support	▪ The following shall be supported as character sets used in messages - UNICODE UTF-8	Support for UNICODE UTF-8 because there is a field to input Japanese Kana character	Fixed
Physical connection			
TCP-IP			
Support	▪ Supported ▪ The number of connections is based on equipment specifications	-	Specification Reference for Each Product
Connection Timeout	▪ Connection timeout shall be set to five seconds and re-connection is not performed ▪ Shall reference use cases in "5.1 Communication Channel Connection/Disconnection"	Required to avoid long user wait times	Specification Reference for Each Product
Ack Receive Timeout	▪ Ack receive timeout shall be set to five seconds and re-send is not performed ▪ Shall reference use cases in "5.2.1.3 Ack Timeout (for Ack Timeout)"	Same as above	Specification Reference for Each Product
Re-send	▪ Maximum re-resend is two times (One time for initial send and one retry). ▪ Shall reference use cases in "5.2.1.2 Receive Ack (for receive Ack)"	Same as above	Fixed
Serial			
Support	▪ Not supported	▪ Because it is not recommended for HL7	-

4. Data Model

Shall indicate the clinical test data model. Areas framed in blue shall be profiles supported in FDC.



5. Transmission/Operation Flow

5.1 Communication Channel Connection/Disconnection

This section describes communication channel connection and disconnection.

In the specimen testing industry, communication channels shall always be connected due to a large volume of permanent connections and no prescription in HL7 protocol.

- After start-up, the connection is started. The connection attempt is repeated until the line is established. After a channel is established, message exchange shall be carried out in the same channel.

However, The message processing unit indicates 1 transaction (=N message) in the IHE integrated profile. A line is established for each transaction type.

- If connection is established, “5.2 Message Flow and Processing” is executed.
- Disconnection is done at shutdown.

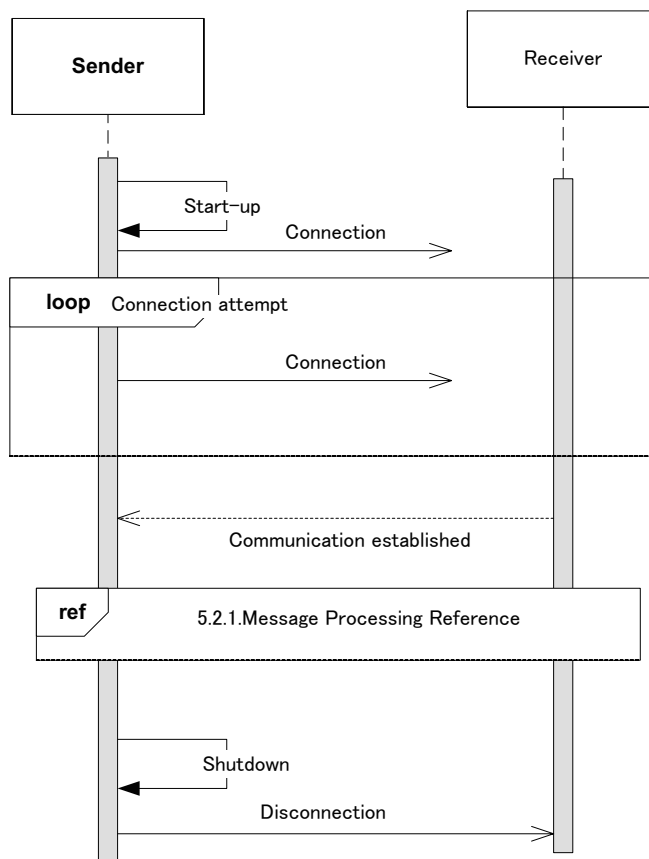


Fig. 5.1-1 Communication Channel Connection/Disconnection Flow

5.2 Message Flow and Processing

Messages shall be processed in accordance with HL7 specification (2.9 Message Processing Rules). Shall receive despite an incongruous version message because HL7 is forward and backward compatible

For forward compatibility:

Supports only known elements and ignores unknown elements

For backward compatibility:

Maintains compatibility for all elements

5.2.1 Send Message

This section compiles processing flow when sending messages.

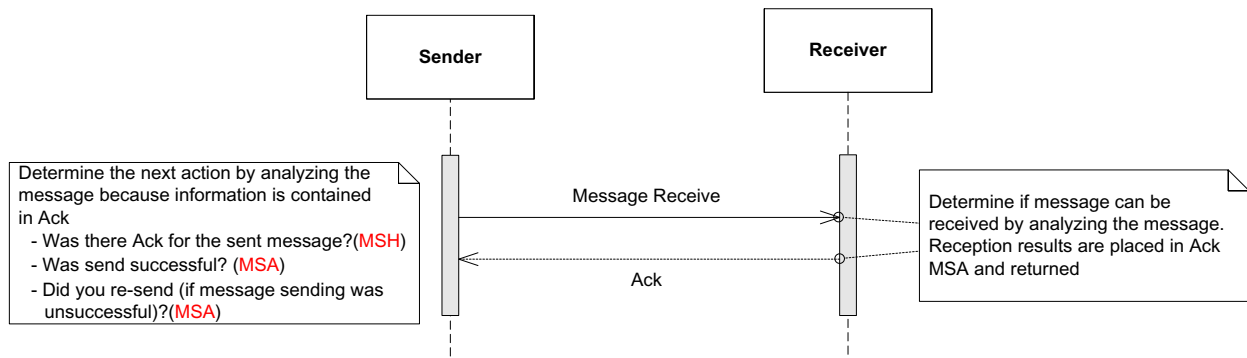
Messages between equipment shall be sent in both directions.

In message send, a message is sent by the Sender, the Receiver analyzes the received message and returns accepted results as Ack. The Sender analyzes the Ack information to determine the subsequent action

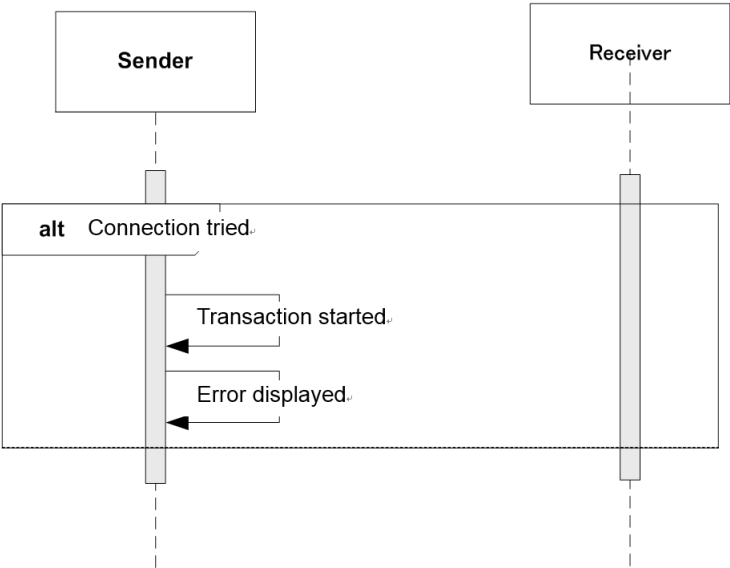
Processing flow is classified as follows and details are compiled in the sections that follow.

- Message receive
- Ack receive
- Ack Timeout
- Message abnormality

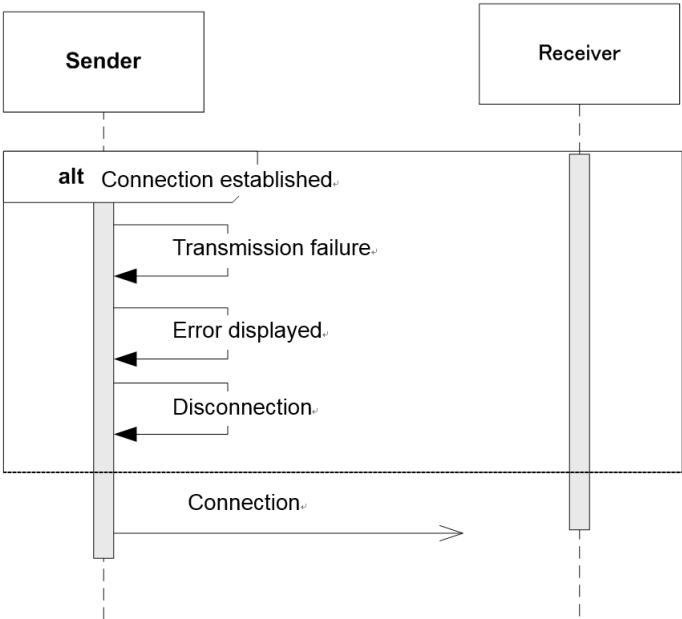
Message content shall reference “6 Message frame specifications”.



When a transaction is started before establishing a line, an error indicating communication abnormality is displayed.



If transmission is failed, reconnection is tried.



5.2.1.1 Receive Message (for receive message)

Occurrence Conditions:

Occurs when the Receiver receives a message

Flow after occurrence:

Message is analyzed in accordance with HL7 specifications

Message received result is returned with ACK regardless of whether it can or cannot be received

5.2.1.2 Receive Ack (for receive Ack)

Occurrence Conditions:

Occurs when the Receiver receives Ack

Flow after occurrence:

Message is processed according to results determination in accordance with HL7 specifications

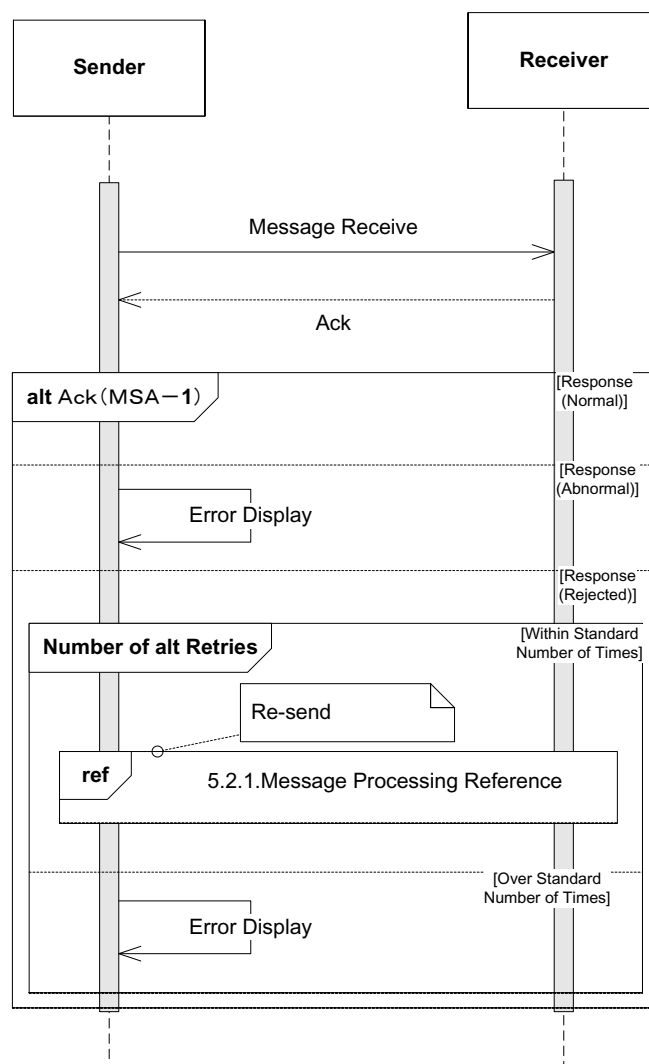
<Results Determination>

Determined with MSA-1 acknowledgement code value.

AA/CA (normal) finishes

AE/CE (abnormal) displays an error and finishes

AR/AR (rejected) message is re-transmitted. When the number of re-sends exceed the standard, displays an error and finishes



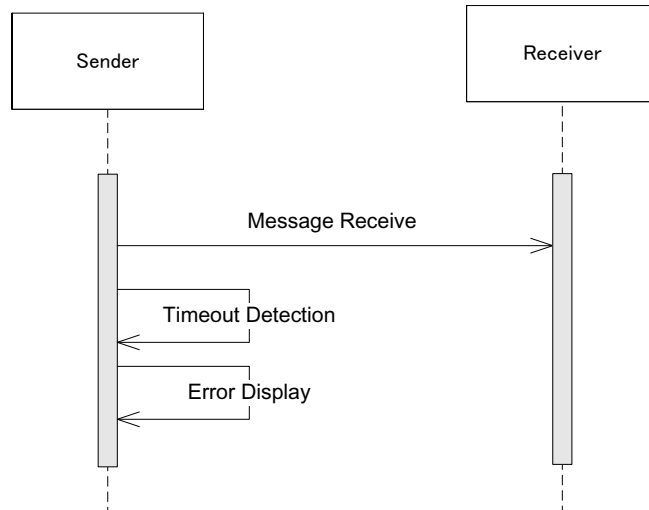
5.2.1.3 Ack Timeout (for Ack Timeout)

Occurrence Conditions:

Occurs when Ack cannot be received by the Sender within the standard value time.

Flow after occurrence:

Displays an error and reconnection is tried



5.2.1.4 Receive unknown Ack (for abnormal Ack message)

Occurrence Conditions:

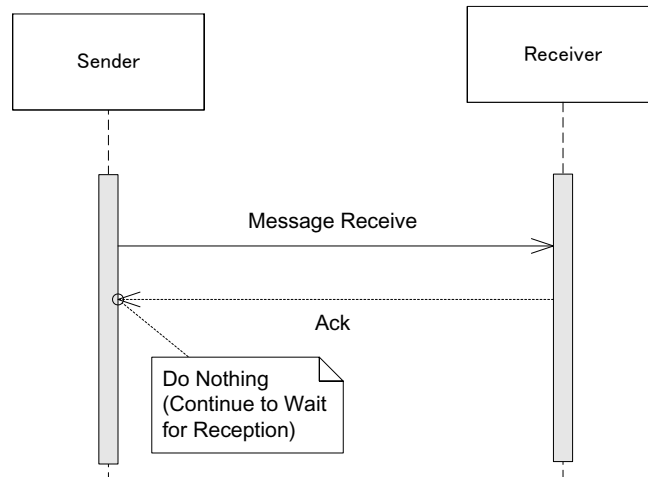
Occurs when the Sender receives Ack but was not the target for reception.

Whether or not it is a reception target is determined within ACK (MSH-5, MSH-6, MSH-11, MSA-2).

Details shall reference each item.

Flow after occurrence:

Ignores message and continues processing



6. Message frame specifications

Messages are prescribed with HL7 specifications for each transaction prescribed in the profile (Refer to “1.5 IHE Profile Support: IHE Integrated Profile Support List”).

This section prescribes message specifications.

Table 6-1 Message List

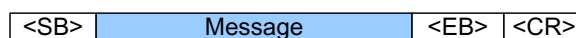
Message Type	Event Code	Description	Sender	Receiver
OUL	R22	OUL - Unsolicited Specimen Oriented Observation Message	FDC	Analyzer Manager
ACK	ALL	ACK - General Acknowledgment message	FDC, Analyzer Manager	FDC, Analyzer Manager

6.1 Overview

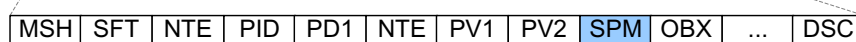
6.1.1 HL7 Message Structure

This section describes HL7 message structure.

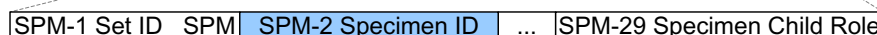
1. Transmission Character Data Structure



2. Message Structure Ex.: Measurement Result (OUL-R22)



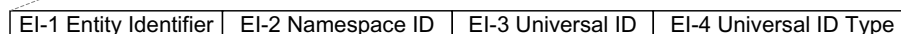
3. Segment Structure Ex.: Specimen Information (SPM)



4. Field Structure Ex.: Specimen ID Information (SPM-2)



5. Component (Data Type) Structure



2

- HL7 message structure is stratified based on the above construction.
- The order is determined: Segment, field, component.
- Data can be abridged to a variable length. Abridging method shall reference each data structure.

6.1.1.1 Transmitted Character Data Structure

- Attached when sending depending on MLLP method settings. Reference “3.2.1 Setting MLLP” for details

6.1.1.2 Message Structure

- Reference “6.2 Unsolicited Specimen Oriented Observation Message (OUL-R22)” and after for each message
- Unsupported segments can be abridged

Ex.: When SFT segment is unsupported

```
MSH|^~\&||||20200108155010||OUL^R22^OUL_R22|1|P|2.5.1||||||LAB-29^IHE  
SFT|  
SPM|1|ID000001||SER^Serum^HL70487||||P^Patient^HL70369
```

The above SFT can be abridged as follows.

```
MSH|^~\&||||20200108155010||OUL^R22^OUL_R22|1|P|2.5.1||||||LAB-29^IHE  
SPM|1|ID000001||SER^Serum^HL70487||||P^Patient^HL70369
```

6.1.1.3 Segment Structure

- Reference “6.2 Unsolicited Specimen Oriented Observation Message (OUL-R22)” and after for each segment
- The identifier between fields is defined by MSH-1 and is normally surrounded by [||]
- Empty unsupported fields

Ex.: When SAC-1, SAC-2 are unsupported

SAC|||123456799

- Unsupported fields after the last used field may be abridged

Ex.: When unsupported after SPM-21

SPM|1|ID000001||SER^Serum^HL70487|||||P^Patient^HL70369||||||||||

The above can be abridged as follows

SPM|1|ID000001||SER^Serum^HL70487|||||P^Patient^HL70369

6.1.1.4 Field Structure

- Reference “6.2 Unsolicited Specimen Oriented Observation Message (OUL-R22)” and after for each field
- The identifier between components is defined by MSH-1 and is normally surrounded by [^]
- In HL7, empty and null are clearly divided and set:

When non-existent, empty (||)

When you wish to send notification of lack of data, null (|""| *2 connected double quotes attached).

- Unsupported components shall be empty (||)

Ex.: When the 1st component is unsupported

|^ID00001|

- Unsupported components after the last used component may be abridged

Ex.: When unsupported after the 2nd component

|ID123456^|

The above can be abridged as follows

|ID123456|

6.1.1.5 Component (Data Type) Structure

- The identifier between subcomponents is defined by MSH-1 and is normally surrounded by [&]
 - In HL7, empty and null are clearly divided and set:
- When non-existent, empty (||),
- When you wish to send notification of lack of data, null (|""| *2 connected double quotes attached).

- Unsupported components shall be empty (^^)

Ex.: When the 1st subcomponent is unsupported

|^ID00001|

- Unsupported subcomponents after the last used subcomponent may be abridged

Ex.: When unsupported after the 2nd subcomponent

^ID123456&^

The above can be abridged as follows

^ID123456^

6.1.2 Legend

The following section and onward describe each item in message, segment and component tables for supported messages.

6.1.2.1 Messages

<Message Table Example>

Segment		Required	FDC Support
MSH	Message Header	R	Yes

- Segment

The left column indicates the abridged name, the right column indicates the segment name. Symbols in the abridged name column are prescribed by HL7 protocol (HL7 Standard 2), [] indicates a segment is optional and {} indicates a segment can loop.

- Required

HL7 output optional. R indicates required and O indicates optional.

- FDC Support

FDC support. Yes: Supports. No: Does not support.

6.1.2.2 Segments

<Segment Table Example>

Sequence	Field Name	Data Type	Repeat	Required	Use by Sender
MSA-1	Acknowledgment Code	ID	1	R	Yes

- Sequence

Field position index in the segment.
Described with [segment abridged name] - [index value].

- Field Name

Field Name

- Data Type

Field Name Data Type

- Repeat

Repeat status. 1: Does not repeat. *: Repeats.

- Required

HL7 output optional. Indications are as follows. R: Required, O: Optional, C:Confidential (conditions attached), X: Not Supported, B: Backward Compatibility, W: Withdrawn.

- Use by Sender or Receiver

FDC output information. Yes: Is output, No: not output

6.1.2.3 Components

<Component Table Example>

■ Component Table: [DT] Type

SEQ	DT	LEN	COMPONENT NAME	Opt	Value	Description
1	ST	20	Identifier	M	<Calculation Code>	▪ Set the code value in OBX-3

▪ SEQ
Component Element Index

▪ DT
Data Type

▪ LEN
Maximum Element Length

▪ COMPONENT NAME
Element Name or Description

▪ Opt
For a sent message, M: Output as required, C: Output with conditions, X: Not output (empty)
For a received message, M: Required (error if no value), X: Unrequired (does not check for value)

▪ Value
Value set with FDC

▪ Description
Supplementary Value Information

6.2 Unsolicited Specimen Oriented Observation Message (OUL-R22)

- Output in this message format (Table 6.2-1) when sending measurement results not ordered as required by AnalyzerManager
- Creates messages per each sample.
- Message sends when measurement ends or re-send is specified.

Table 6.2-1 OUL-R22 Message

Segment		Required	FDC Support
MSH	Message Header segment	R	Yes
[[SFT]]	Software Segment	O	No
[NTE]	Notes and Comments segment	O	No
[	-- PATIENT begin	O	No
PID	Patient Identification segment	R	No
[PD1]	Patient Additional Demographic segment	O	No
[[NTE]]	Notes and Comments segment	O	No
]	-- PATIENT end	-	-
[	-- VISIT begin	O	No
PV1	Patient Visit segment	R	No
[PV2]	Patient Visit - Additional Information Segment	O	No
]	-- VISIT end	-	-
{	-- SPECIMEN begin	R	Yes
SPM	Specimen information segment	R	Yes
[[OBX]]	Observation Result(for Specimen) Segment	O	No
[[	-- CONTAINER begin	O	Yes
SAC	Container information	R	Yes
[INV]	Detailed Substance information segment	O	No
]]	-- CONTAINER end	-	-
{	-- ORDER begin	R	Yes
OBR	Observation Order segment	R	Yes
[ORC]	Common Order segment	O	Yes
[[NTE]]	Notes and Comments segment	O	No
[[	-- TIMING_QTY begin	O	No
TQ1	Timing/Quantity segment	R	No
[[TQ2]]	Timing/Quantity Relationship segment	O	No
]]	-- TIMING_QTY end	-	-
[[	-- RESULT begin	O	Yes
OBX	Observation Result segment	R	Yes
[TCD]	Test Code Detail segment	O	Yes
[[SID]]	Substance Identifier segment	O	No
[[NTE]]	Notes and Comments segment	O	No
]]	-- RESULT end	-	-
[[CTI]]	Clinical Trial Identification segment	O	No
}	-- ORDER end	-	-
}	-- SPECIMEN end	-	-
[DSC]	Continuation Pointer segment	O	No

6.2.1 Used Segment

6.2.1.1 MSH-Message Header Segment

Table 6.2.1-1 MSH Segment

Sequence	Field Name	Data Type	Repeat	Required	Use by Sender Or Receiver
MSH-1	Field Separator	ST	1	R	Yes
MSH-2	Encoding Characters	ST	1	R	Yes
MSH-3	Sending Application	HD	1	O	No
MSH-4	Sending Facility	HD	1	O	No
MSH-5	Receiving Application	HD	1	O	Yes(if exists)
MSH-6	Receiving Facility	HD	1	O	Yes(if exists)
MSH-7	Date/Time Of Message	TS	1	R	Yes
MSH-8	Security	ST	1	O	No
MSH-9	Message Type	MSG	1	R	Yes
MSH-10	Message Control ID	ST	1	R	Yes
MSH-11	Processing ID	PT	1	R	Yes
MSH-12	Version ID	VID	1	R	Yes
MSH-13	Sequence Number	NM	1	O	No
MSH-14	Continuation Pointer	ST	1	O	No
MSH-15	Accept Acknowledgment Type	ID	1	O	No
MSH-16	Application Acknowledgment Type	ID	1	O	No
MSH-17	Country Code	ID	1	O	No
MSH-18	Character Set	ID	*	O	Yes
MSH-19	Principal Language Of Message	CE	1	O	No
MSH-20	Alternate Character Set Handling Scheme	ID	1	O	No
MSH-21	Message Profile Identifier	EI	*	O	Yes

MSH-1 Field Separator

Set field separator characters. Set recommended value "|" (ASCII 124).

■ Component table: ST type (String Data)

SEQ	DT	LEN	COMPONENT NAME	Opt	Value	Description
-	-	199	String Data	M	<Field Separator Characters>	Set

MSH-2 Encoding Characters

Set coding characters (Component Separator, Repetition Separator, Escape Separator, Subcomponent Separator). Set recommended values "^" (ASCII 94), "~" (ASCII 126), "\" (ASCII 92), "&" (ASCII 38).

■ Component table: ST type (String Data)

SEQ	DT	LEN	COMPONENT NAME	Opt	Value	Description
-	-	199	String Data	M	<Coding Characters>	Set ^~\&

MSH-5 Receiving Application

■ Component Table: HD Type

SEQ	DT	LEN	COMPONENT NAME	Opt	Value	Description
1	IS	20	Namespace ID	M	<Received Application ID>	Set (" "(null)) when there is no data
2	ST	199	Universal ID	X		Unsupported
3	ID	6	Universal ID Type	X		Unsupported

MSH-6 Receiving Facility

■ Component Table: HD Type

SEQ	DT	LEN	COMPONENT NAME	Opt	Value	Description
1	IS	20	Namespace ID	M	<Receiver>	Set (" "(null)) when there is no data
2	ST	199	Universal ID	X		Unsupported
3	ID	6	Universal ID Type	X		Unsupported

MSH-7 Date/Time Of Message

Set the message creation date/time (seconds)

■ Component Table: TS Type

SEQ	DT	LEN	COMPONENT NAME	Opt	Value	Description
1	DTM	24	Time	M	<Message Creation Date/ Time>	Set in <YYYYMMDDHHMMSS> format
2	ID	1	Degree of Precision	X		Unsupported

MSH-9 Message Type

Set message format according to the created message (See Table 5-1)

■ Component table: MSG Type

SEQ	DT	LEN	COMPONENT NAME	Opt	Value	Description
1	ID	3	Message Code	M	<Message Code Value>	Ex.: For OUL-R22 OUL
2	ID	3	Trigger Event	M	<Trigger Event>	Ex.: For OUL-R22 R22
3	ID	7	Message Structure	M	<Message Structure>	Ex.: For OUL-R22 OUL_R22

MSH-10 Message Control ID

Set the message control ID (integers starting from 1).

Set to 1 upon equipment startup, increments with each created message.

■ Componen Table: ST type (String Data)

SEQ	DT	LEN	COMPONENT NAME	Opt	Value	Description
-	-	199	String Data	M	<Message Control ID>	

MSH-11 Processing ID

Set the processing ID. P: Set production.

■ Component Table: PT Type

SEQ	DT	LEN	COMPONENT NAME	Opt	Value	Description
1	ID	1	Processing ID	M	<Processing ID>	Set "P" (fixed)
2	ID	1	Processing Mode	X	<Processing Mode>	Unsupported

MSH-12 Version ID

Set the HL7 version to be used. Set 2.5.1 to adopt v2.5.1.

■ Component Table: VID Type

SEQ	DT	LEN	COMPONENT NAME	Opt	Value	Description
1	ID	5	Version ID	M	<HL7 Version>	Set 2.5.1 (fixed)
2	CE	483	Internationalization Code	X		Unsupported
3	CE	483	International Version ID	X		Unsupported

MSH-18 Character Set

Because Japanese Kana character is sometimes outputted, UNICODE UTF-8 is set.

■ Component table: ID Type(String Data)

SEQ	DT	LEN	COMPONENT NAME	Opt	Value	Description
-	-	199	Character Set	M	<Caracter set>	「UNICODE UTF-8」 (fixed) is set

MSH-21 Message Profile Identifier

Set the identifier for the message profile to be created.

■ Component Table: EI Type

SEQ	DT	LEN	COMPONENT NAME	Opt	Value	Description
1	ST	199	Entity Identifier	M	<Message Profile ID>	LAB-29 for OUL-R22
2	IS	20	Namespace ID	M	<Message Profile System>	Set "IHE" (fixed)
3	ST	199	Universal ID	X		Unsupported
4	ID	6	Universal ID Type	X		Unsupported

6.2.1.2 SPM-Specimen information Segment

Table 6.2.1.2-1 Specimen information Segment

Sequence	Field Name	Data Type	Repeat	Required	Use by Sender
SPM-1	Set ID _ SPM	SI	1	O	Yes
SPM-2	Specimen ID	EIP	1	O	Yes
SPM-3	Specimen Parent IDs	EIP	*	O	No
SPM-4	Specimen Type	CWE	1	R	Yes
SPM-5	Specimen Type Modifier	CWE	*	O	No
SPM-6	Specimen Additives	CWE	*	O	No
SPM-7	Specimen Collection Method	CWE	1	O	No
SPM-8	Specimen Source Site	CWE	1	O	No
SPM-9	Specimen Source Site Modifier	CWE	*	O	No
SPM-10	Specimen Collection Site	CWE	1	O	No
SPM-11	Specimen Role	CWE	*	O	Yes
SPM-12	Specimen Collection Amount	CQ	1	O	No
SPM-13	Grouped Specimen Count	NM	1	C	No
SPM-14	Specimen Description	ST	*	O	No
SPM-15	Specimen Handling Code	CWE	*	O	No
SPM-16	Specimen Risk Code	CWE	*	O	No
SPM-17	Specimen Collection Date/Time	DR	1	O	No
SPM-18	Specimen Received Date/Time	TS	1	O	No
SPM-19	Specimen Expiration Date/Time	TS	1	O	No
SPM-20	Specimen Availability	ID	1	O	No
SPM-21	Specimen Reject Reason	CWE	*	O	No
SPM-22	Specimen Quality	CWE	1	O	No
SPM-23	Specimen Appropriateness	CWE	1	O	No
SPM-24	Specimen Condition	CWE	*	O	No
SPM-25	Specimen Current Quantity	CQ	1	O	No
SPM-26	Number of Specimen Containers	NM	1	O	No
SPM-27	Container Type	CWE	1	O	No
SPM-28	Container Condition	CWE	1	O	No
SPM-29	Specimen Child Role	CWE	1	O	No

SPM-1 Set ID _ SPM

Set 1 (fixed) for single specimen results

■ Component Table: SI Type

SEQ	DT	LEN	COMPONENT NAME	Opt	Value	Description
-	-	4	Sequence ID	M	<Sequence ID>	▪ Set 1 (fixed)

SPM-2 Specimen ID

Set the results screen specimen ID. The specimen ID is information that can be input by the user during specimen information registration (" " (null) included).

■ Component Table: EIP Type

SEQ	DT	LEN	COMPONENT NAME	Opt	Value	Description
1	EI	427	Placer Assigned Identifier	X		Unsupported
2	EI	427	Filler Assigned Identifier	M	-	-
2.1	ST	199	Entity Identifier	M	<Specimen ID>	▪ Japanese Kana character may be set (when Japanese Kana character is entered in sample IDs)
2.2	IS	20	Namespace ID	X		Unsupported
2.3	ST	199	Universal ID	X		Unsupported
2.4	ID	6	Universal ID Type	X		Unsupported

SPM-4 Specimen Type

Set the results screen specimen type. Use HL7table0487.

■ Component Table: CWE Type

SEQ	DT	LEN	COMPONENT NAME	Opt	Value	Description
1	ST	20	Identifier	M	<Specimen Type Code Value>	▪ Set the code value in Table SPM-4-1
2	ST	199	Text	C	<Specimen Type Description>	▪ Set description in Table SPM-4-1
3	ID	20	Name of Coding System	C	<Specimen Type Code System>	▪ Reference the code system in Table SPM-4-1
4	ST	20	Alternate Identifier	X		Unsupported
5	ST	199	Alternate Text	X		Unsupported
6	ID	20	Name of Alternate Coding System	X		Unsupported
7	ST	10	Coding System Version ID	X		Unsupported
8	ST	10	Alternate Coding System Version ID	X		Unsupported
9	ST	199	Original Text	X		Unsupported

2

Table SPM-4-1 Specimen Type Code

PF or IC	By Measurement Type	Specimen Type	Value		
			Code Value	Description	Code System
Yes	Automatic/Manual Spot Application Measurement	Plasma/Blood Serum	PLAS	Plasma	HL70487
No	Automatic/Manual Spot Application Measurement	Plasma *	PLAS	Plasma	HL70487
		Blood Serum *	SER	Serum	HL70487
		Whole Blood	WB	Blood, Whole	↑
		Urine	UR	Urine	↑
	Control Measurement	Blood Serum/Plasma	" "(null)		
		Whole Blood			

Control measurement must be set as " "(null)" in HL7 specifications.

* Selection of Plasma/Serum is done by type 4 setting of mode function.

SPM-11 Specimen Role

This field is for setting the specimen role, but the specimen role is associated by measurement type in the Analyzer.

Use HL7table0369.

■ Component Table: CWE Type

SEQ	DT	LEN	COMPONENT NAME	Opt	Value	Description
1	ST	20	Identifier	M	<Specimen Role Code Value>	▪ Set the code value in Table SPM-11-1
2	ST	199	Text	C	<Specimen Role Description>	▪ Set description in Table SPM-11-1
3	ID	20	Name of Coding System	C	<Specimen Role Code System>	▪ Reference the code system in Table SPM-11-1
4	ST	20	Alternate Identifier	X		Unsupported
5	ST	199	Alternate Text	X		Unsupported
6	ID	20	Name of Alternate Coding System	X		Unsupported
7	ST	10	Coding System Version ID	X		Unsupported
8	ST	10	Alternate Coding System Version ID	X		Unsupported
9	ST	199	Original Text	X		Unsupported

Table SPM-11-1 Code by Measurement Type

By Measurement Type	Value		
	Code Value	Description	Code System
Automatic/Manual Spot Application Measurement	P	Patient	HL70369
Control Measurement	Q	Control specimen	HL70369

6.2.1.3 SAC-Container information Segment

Table 6.2.1.3-1 Container information Segment

Sequence	Field Name	Data Type	Repeat	Required	Use by Sender
SAC-1	External Accession Identifier	EI	1	O	No
SAC-2	Accession Identifier	EI	1	O	No
SAC-3	Container Identifier	EI	1	C	Yes
SAC-4	Primary	EI	1	C	No
SAC-5	Equipment Container Identifier	EI	1	O	No
SAC-6	Specimen Source	SPS	1	C	No
SAC-7	Registration Date/Time	TS	1	O	No
SAC-8	Container Status	CE	1	O	No
SAC-9	Carrier Type	CE	1	O	No
SAC-10	Carrier Identifier	EI	1	O	No
SAC-11	Position in Carrier	NA	1	O	No
SAC-12	Tray Type - SAC	CE	1	O	No
SAC-13	Tray Identifier	EI	1	O	No
SAC-14	Position in Tray	NA	1	O	No
SAC-15	Location	CE	*	O	Yes
SAC-16	Container Height	NM	1	O	No
SAC-17	Container Diameter	NM	1	O	No
SAC-18	Barrier Delta	NM	1	O	No
SAC-19	Bottom Delta	NM	1	O	No
SAC-20	Container Height/Diameter/Delta Units	CE	1	O	No
SAC-21	Container Volume	NM	1	O	No
SAC-22	Available Specimen Volume	NM	1	O	No
SAC-23	Initial Specimen Volume	NM	1	O	No
SAC-24	Volume Units	CE	1	O	No
SAC-25	Separator Type	CE	1	O	No
SAC-26	Cap Type	CE	1	O	No
SAC-27	Additive	CWE	*	O	No
SAC-28	Specimen Component	CE	1	O	No
SAC-29	Dilution Factor	SN	1	O	No
SAC-30	Treatment	CE	1	O	No
SAC-31	Temperature	SN	1	O	No
SAC-32	Hemolysis Index	NM	1	O	No
SAC-33	Hemolysis Index Units	CE	1	O	No
SAC-34	Lipemia Index	NM	1	O	No
SAC-35	Lipemia Index Units	CE	1	O	No
SAC-36	Icterus Index	NM	1	O	No
SAC-37	Icterus Index Units	CE	1	O	No
SAC-38	Fibrin Index	NM	1	O	No
SAC-39	Fibrin Index Units	CE	1	O	No
SAC-40	System Induced Contaminants	CE	*	O	No
SAC-41	Drug Interference	CE	*	O	No
SAC-42	Artificial Blood	CE	1	O	No
SAC-43	Special Handling Code	CWE	*	O	No
SAC-44	Other Environmental Factors	CE	*	O	No

SAC-3 Container Identifier

Set the specimen container ID number. Set the No. in the results screen because the specimen ID number cannot be recognized in the Analyzer. The No. automatically increments when specimen information is registered, and information can be edited by the user (integer of 1 or more).

■ Component Table: EI Type

SEQ	DT	LEN	COMPONENT NAME	Opt	Value	Description
1	ST	199	Entity Identifier	M	<No>	-
2	IS	20	Namespace ID	X		Unsupported
3	ST	199	Universal ID	X		Unsupported
4	ID	6	Universal ID Type	X		Unsupported

SAC-15 Location

Set the specimen container location. In the Analyzer, set the position where the specimen is placed

■ Component Table: CE Type

SEQ	DT	LEN	COMPONENT NAME	Opt	Value	Description
1	ST	20	Identifier	M	<Specimen Position Code>	▪ Set the specimen position index based on each product specification. Ex.: 1
2	ST	199	Text	M	<Specimen Position Description>	▪ Enter position description Ex.: A
3	ID	20	Name of Coding System	M	<Specimen Position System>	▪ Set "L"
4	ST	20	Alternate Identifier	X		Unsupported
5	ST	199	Alternate Text	X		Unsupported
6	ID	20	Name of Alternate Coding System	X		Unsupported

6.2.1.4 OBR-Observation Order Segment

Table 6.2.1.4-1 Observation Order Segment

Sequence	Field Name	Data Type	Repeat	Required	Use by Sender
OBR-1	Set ID - OBR	SI	1	O	No
OBR-2	Placer Order Number	EI	1	C	No
OBR-3	Filler Order Number	EI	1	C	No
OBR-4	Universal Service Identifier	CE	1	R	Yes
OBR-5	Priority _ OBR	ID	1	B	No
OBR-6	Requested Date/Time	TS	1	B	No
OBR-7	Observation Date/Time	TS	1	C	No
OBR-8	Observation End Date/Time	TS	1	O	No
OBR-9	Collection Volume	CQ	1	O	No
OBR-10	Collector Identifier	XCN	*	O	No
OBR-11	Specimen Action Code	ID	1	O	No
OBR-12	Danger Code	CE	1	O	No
OBR-13	Relevant Clinical Information	ST	1	O	No
OBR-14	Specimen Received Date/Time	TS	1	B	No
OBR-15	Specimen Source	SPS	1	B	No
OBR-16	Ordering Provider	XCN	*	O	No
OBR-17	Order Callback Phone Number	XTN	*	O	No
OBR-18	Placer Field 1	ST	1	O	No
OBR-19	Placer Field 2	ST	1	O	No
OBR-20	Filler Field 1	ST	1	O	No
OBR-21	Filler Field 2	ST	1	O	No
OBR-22	Results Rpt/Status Chng - Date/Time	TS	1	C	No
OBR-23	Charge to Practice	MOC	1	O	No
OBR-24	Diagnostic Serv Sect ID	ID	1	O	No
OBR-25	Result Status	ID	1	C	No
OBR-26	Parent Result	PRL	1	O	No
OBR-27	Quantity/Timing	TQ	*	B	No
OBR-28	Result Copies To	XCN	*	O	No
OBR-29	Parent Number	EIP	1	O	No
OBR-30	Transportation Mode	ID	1	O	No
OBR-31	Reason for Study	CE	*	O	No
OBR-32	Principal Result Interpreter	NDL	1	O	No
OBR-33	Assistant Result Interpreter	NDL	*	O	No
OBR-34	Technician	NDL	*	O	No
OBR-35	Transcriptionist	NDL	*	O	No
OBR-36	Scheduled Date/Time	TS	1	O	No
OBR-37	Number of Sample Containers *	NM	1	O	No
OBR-38	Transport Logistics of Collected Sample	CE	*	O	No
OBR-39	Collector's Comment *	CE	*	O	No
OBR-40	Transport Arrangement Responsibility	CE	1	O	No
OBR-41	Transport Arranged	ID	1	O	No
OBR-42	Escort Required	ID	1	O	No
OBR-43	Planned Patient Transport Comment	CE	*	O	No
OBR-44	Procedure Code	CE	1	O	No
OBR-45	Procedure Code Modifier	CE	*	O	No
OBR-46	Placer Supplemental Service Information	CE	*	O	No
OBR-47	Filler Supplemental Service Information	CE	*	O	No
OBR-48	Medically Necessary Duplicate Procedure Reason.	CWE	1	C	No
OBR-49	Result Handling	IS	1	O	No
OBR-50	Parent Universal Service Identifier	CWE	1	O	No

OBR-4 Universal Service Identifier

Set test item ID.

Define as unclear because the Analyzer does not hold test information

■ Component Table: CE Type

SEQ	DT	LEN	COMPONENT NAME	Opt	Value	Description
1	ST	20	Identifier	M	<Test Code>	▪ Set "U" (fixed)
2	ST	199	Text	M	<Test Description>	▪ Set "Unspecified identifier" (fixed)
3	ID	20	Name of Coding System	M	<Test Code System>	▪ Set "L" (fixed)
4	ST	20	Alternate Identifier	X		Unsupported
5	ST	199	Alternate Text	X		Unsupported
6	ID	20	Name of Alternate Coding System	X		Unsupported

6.2.1.5 ORC-Common Order Segment

Table 6.2.1.5-1 Common Order Segment

Sequence	Field Name	Data Type	Repeat	Required	Use by Sender
ORC-1	Order Control	ID	1	R	Yes
ORC-2	Placer Order Number	EI	1	C	No
ORC-3	Filler Order Number	EI	1	C	No
ORC-4	Placer Group Number	EI	1	O	No
ORC-5	Order Status	ID	1	O	Yes
ORC-6	Response Flag	ID	1	O	No
ORC-7	Quantity/Timing	TQ	*	B	No
ORC-8	Parent Order	EIP	1	O	No
ORC-9	Date/Time of Transaction	TS	1	O	No
ORC-10	Entered By	XCN	*	O	No
ORC-11	Verified By	XCN	*	O	No
ORC-12	Ordering Provider	XCN	*	O	No
ORC-13	Enterer's Location	PL	1	O	No
ORC-14	Call Back Phone Number	XTN	*	O	No
ORC-15	Order Effective Date/Time	TS	1	O	No
ORC-16	Order Control Code Reason	CE	1	O	No
ORC-17	Entering Organization	CE	1	O	No
ORC-18	Entering Device	CE	1	O	No
ORC-19	Action By	XCN	*	O	No
ORC-20	Advanced Beneficiary Notice Code	CE	1	O	No
ORC-21	Ordering Facility Name	XON	*	O	No
ORC-22	Ordering Facility Address	XAD	*	O	No
ORC-23	Ordering Facility Phone Number	XTN	*	O	No
ORC-24	Ordering Provider Address	XAD	*	O	No
ORC-25	Order Status Modifier	CWE	1	O	No
ORC-26	Advanced Beneficiary Notice Override Reason	CWE	1	C	No
ORC-27	Filler's Expected Availability Date/Time	TS	1	O	No
ORC-28	Confidentiality Code	CWE	1	O	No
ORC-29	Order Type	CWE	1	O	No
ORC-30	Enterer Authorization Mode	CNE	1	O	No
ORC-31	Parent Universal Service Identifier	CWE	1	O	No

ORC-1 Order Control

Set order control. Set SC: Status Changed (fixed) from IHE LAB-29

■ Component table: ID type (String Data)

SEQ	DT	LEN	COMPONENT NAME	Opt	Value	Description
-	-	199	Coded Value for HL7-Defined Tables	M	<Order Control>	▪ Set "SC" (fixed)

ORC-5 Order Status

Set test status. Set CM: Complete (fixed) because it is only output after test is complete.

■ Component table: ID type (String Data)

SEQ	DT	LEN	COMPONENT NAME	Opt	Value	Description
-	-	199	Coded Value for HL7-Defined Tables	M	<Test Status>	▪ Set "CM" (fixed)

6.2.1.6 OBX-Observation Result Segment

6.2.1.6.1 Observation Result Segment

Table 6.2.1.6-1 Observation Result Segment

Sequence	Field Name	Data Type	Repeat	Required	Use by Sender
OBX-1	Set ID - OBX	SI	1	O	Yes
OBX-2	Value Type	ID	125	C	Yes
OBX-3	Observation Identifier	CE	1	R	Yes
OBX-4	Observation Sub-ID	ST	1	C	No
OBX-5	Observation Value	VARIES	*	C	Yes
OBX-6	Units	CE	1	O	Yes
OBX-7	References Range	ST	1	O	Yes
OBX-8	Abnormal Flags	IS	*	O	Yes
OBX-9	Probability	NM	1	O	No
OBX-10	Nature of Abnormal Test	ID	*	O	No
OBX-11	Observation Result Status	ID	1	R	Yes
OBX-12	Effective Date of Reference Range	TS	1	O	No
OBX-13	User Defined Access Checks	ST	1	O	No
OBX-14	Date/Time of the Observation	TS	1	O	No
OBX-15	Producer's ID	CE	1	O	No
OBX-16	Responsible Observer	XCN	*	O	Yes
OBX-17	Observation Method	CE	*	O	No
OBX-18	Equipment Instance Identifier	EI	*	O	Yes
OBX-19	Date/Time of the Analysis	TS	1	O	Yes
OBX-20	Reserved for v2.6	ST	1	O	No
OBX-21	Reserved for v2.6	ST	1	O	No
OBX-22	Reserved for v2.6	ST	1	O	No
OBX-23	Performing Organization Name	XON	1	O	No
OBX-24	Performing Organization Address	XAD	1	O	No
OBX-25	Performing Organization Medical Director	XCN	1	O	No

OBX-1 Set ID - OBX

Set the OBX index value (integers starting from 1). Increments each repeat of OBX

■ Component table: SI type (Numeric Data)

SEQ	DT	LEN	COMPONENT NAME	Opt	Value	Description
-	-	4	Sequence ID	M	<OBX Repeat Index Value>	▪ Increments each repeat of OBX

OBX-2 Value Type

Specify Data Type according to the value of single item/between item calculation.

■ Component table: ID type (String Data)

SEQ	DT	LEN	COMPONENT NAME	Opt	Value	Description
-	-	20	Coded Value for HL7-Defined Tables	M	<Calculated Value Data Type>	▪ Set "NM" (fixed)

OBX-3 Observation Identifier

Set the single item/between item calculation displayed on the results screen according to various codes.

■ Component Table: CE Type

SEQ	DT	LEN	COMPONENT NAME	Opt	Value	Description
1	ST	20	Identifier	M	<Calculation Code>	▪ Set "LOINC Code" in Appendix 2
2	ST	199	Text	M	<Calculation Description>	▪ Set "Sample Type Code" in Appendix 2
3	ID	20	Name of Coding System	M	<Calculation Code System>	▪ LOINC: LN ▪ Locally defined: L
4	ST	20	Alternate Identifier	X		Unsupported
5	ST	199	Alternate Text	X		Unsupported
6	ID	20	Name of Alternate Coding System	X		Unsupported

Appendix 2 FDC HL7 sample type code (LOINC)

OBX-5 Observation Value

Set calculation results value.

■ Component Table: VALUES Type

SEQ	DT	LEN	COMPONENT NAME	Opt	Value	Description
-	-	-	String Data	M	<Calculation Value>	▪ Set “ ” (null) when no calculated value is produced

OBX-6 Units

Set unit code using UCUM code.

■ Component Table: CE Type

SEQ	DT	LEN	COMPONENT NAME	Opt	Value	Description
1	ST	20	Identifier	M	<Unit Code Value>	▪ Set the code value in Table OBX-6-1 when there is a unit to set ▪ Set “ ” (null) when there is not a unit to set
2	ST	199	Text	X		
3	ID	20	Name of Coding System	C	<Unit Code System>	▪ Set “UCUM” or “L” when there is a unit to set ▪ Do not output when there is not a unit
4	ST	20	Alternate Identifier	X		Unsupported
5	ST	199	Alternate Text	X		Unsupported
6	ID	20	Name of Alternate Coding System	X		Unsupported

Table OBX-6-1 Unit Code

Unit	Code Value	Notes
mg/dl	mg/dL	
mmol/l	mmol/L	
umol/l	umol/L	
U/l	U/L	
K-U	—	Output locally defined (“K-U^^L”) because it is not defined in UCUM
WL-U	—	Output locally defined (“WL-U^^L”) because it is not defined in UCUM
KA-U	[ka’U]	
S-U	[smgy’U]	
g/l	g/L	
g/dl	g/dL	
mg/l	mg/L	
ug/dl	ug/dL	
mEq/l	meq/L	
ukat/l	ukat/L	
mg/mg	—	Output locally defined (“mg/mg ^^L”) because it is not defined in UCUM
(SI)	—	Output locally defined (“(SI)^^L”) because it is not defined in UCUM

OBX-7 References Range

Set the reference standard value

■ Component table: ST type (String Data)

SEQ	DT	LEN	COMPONENT NAME	Opt	Value	Description
-	-	199	String Data	M	<Reference Standard Value Minimum> - <Reference Standard Value Maximum>	▪ Set " " (null) when there is not a reference standard value ▪ Set a reference standard range for automatic/manual spot application measurement. ▪ Set a control solution range for control measurement

OBX-8 Abnormal Flags

Set the abnormal flag according to error status.

■ Component Table: IS Type (String Data)

SEQ	DT	LEN	COMPONENT NAME	Opt	Value	Description
-	-	20	Coded Value for User-Defined Tables	M	<Abnormal Flags>	▪ Set the Table OBX-8-1 abnormal flag value

OBX-8-1 Abnormal Flags

In normal JAHIS, setting " "(space) is recommended, but this is not defined in HL7/IHE.

Set " " (null) because an undefined value may not be accepted and this field is for setting abnormal status.

See "8 Appendix 3 Relationship between Measurement Results and OBX-2, 5, 8, 11" for the relationship between measurement results (re-print data) and OBX-2, 5, 8 and 11.

OBX-11 Observation Result Status

Set result status according to value calculation status.

■ Component table: ID type (String Data)

SEQ	DT	LEN	COMPONENT NAME	Opt	Value	Description
-	-	199	Coded Value for User-Defined Tables	M	<Abnormal Flags>	▪ Set the Table OBX-11-1 abnormal flag value

Table OBX-11-1 Result Status

Value Calculation Present	Value
Yes	F
No	X

OBX-16 Responsible Observer

Set the person in charge of testing. Set the operator ID in consideration of the person who executes and verifies testing in the Analyzer.

■ Component Table: XCN Type

SEQ	DT	LEN	COMPONENT NAME	Opt	Value	Description
1	ST	15	ID Number	M	<Operator ID>	▪ Set “ ” (null) when there is not an operator ID ▪ Japanese Kana character may be set (when Japanese Kana character is entered in operator IDs)
2	FN	194	Family Name	X		Unsupported
3	ST	30	Given Name	X		Unsupported
4	ST	30	Second and Further Given Names or Initials Thereof	X		Unsupported
5	ST	20	Suffix (e.g., JR or III)	X		Unsupported
6	ST	20	Prefix (e.g., DR)	X		Unsupported
7	IS	5	Degree (e.g., MD)	X		Unsupported
8	IS	4	Source Table	X		Unsupported
9	HD	227	Assigning Authority	X		Unsupported
10	ID	1	Name Type Code	X		Unsupported
11	ST	1	Identifier Check Digit	X		Unsupported
12	ID	3	Check Digit Scheme	X		Unsupported
13	ID	5	Identifier Type Code	X		Unsupported
14	HD	227	Assigning Facility	X		Unsupported
15	ID	1	Name Representation Code	X		Unsupported
16	CE	483	Name Context	X		Unsupported
17	DR	53	Name Validity Range	X		Unsupported
18	ID	1	Name Assembly Order	X		Unsupported
19	TS	26	Effective Date	X		Unsupported
20	TS	26	Expiration Date	X		Unsupported
21	ST	199	Professional Suffix	X		Unsupported
22	CWE	705	Assigning Jurisdiction	X		Unsupported
23	CWE	705	Assigning Agency or Department	X		Unsupported

OBX-18 Equipment Instance Identifier

Set the equipment serial number

■ Component Table: EI Type

SEQ	DT	LEN	COMPONENT NAME	Opt	Value	Description
1	ST	199	Entity Identifier	M	<S/N>	-
2	IS	20	Namespace ID	X		Unsupported
3	ST	199	Universal ID	X		Unsupported
4	ID	6	Universal ID Type	X		Unsupported

OBX-19 Date/Time of the Analysis

Set the measurement complete date/time. In the Analyzer, set the measurement date and time of the results screen (end date and time of first spot application measurement).

■ Component Table: TS Type

SEQ	DT	LEN	COMPONENT NAME	Opt	Value	Description
1	DTM	24	Time	M	<Measurement Date/Time>	▪ Format is <YYYYMMDDHHMMSS>
2	ID	1	Degree of Precision	X		Unsupported

6.2.1.6.2 Use MDT Option

The added information "Microscopic Or Secondary Description" of OBX is used and written to add the information to the result of 6.2.1.6.1.

Refer to "6.2.2 Example Messages" for the sample example of the message.

Table 6.2.1.6-1 Observation Result Segment

Sequence	Field Name	Data Type	Repeat	Required	Use by Sender
OBX-1	Set ID - OBX	SI	1	O	Yes
OBX-2	Value Type	ID	125	C	Yes
OBX-3	Observation Identifier	CE	1	R	Yes
OBX-4	Observation Sub-ID	ST	1	C	No
OBX-5	Observation Value	VARIES	*	C	Yes
OBX-6	Units	CE	1	O	No
OBX-7	References Range	ST	1	O	No
OBX-8	Abnormal Flags	IS	*	O	No
OBX-9	Probability	NM	1	O	No
OBX-10	Nature of Abnormal Test	ID	*	O	No
OBX-11	Observation Result Status	ID	1	R	No
OBX-12	Effective Date of Reference Range	TS	1	O	No
OBX-13	User Defined Access Checks	ST	1	O	No
OBX-14	Date/Time of the Observation	TS	1	O	No
OBX-15	Producer's ID	CE	1	O	No
OBX-16	Responsible Observer	XCN	*	O	No
OBX-17	Observation Method	CE	*	O	No
OBX-18	Equipment Instance Identifier	EI	*	O	No
OBX-19	Date/Time of the Analysis	TS	1	O	No
OBX-20	Reserved for v2.6	ST	1	O	No
OBX-21	Reserved for v2.6	ST	1	O	No
OBX-22	Reserved for v2.6	ST	1	O	No
OBX-23	Performing Organization Name	XON	1	O	No
OBX-24	Performing Organization Address	XAD	1	O	No
OBX-25	Performing Organization Medical Director	XCN	1	O	No

OBX-1 Set ID - OBX

Same as the OBX-1 of 6.2.1.6.1

OBX-2 Value Type

■ Component table : ID Type (String Data)

SEQ	DT	LEN	COMPONENT NAME	Opt	Value	Description
-	-	20	Coded Value for HL7-Defined Tables	M	<Added Information Data Type>	▪ Set "TX" (fixed)

OBX-3 Observation Identifier

An MDT representing added information is added to the OBX-3 in 6.2.1.6.1

■Component table : CE Type

SEQ	DT	LEN	COMPONENT NAME	Opt	Value	Description
1	ST	20	Identifier	M	<Calculation Code>&MDT	▪ Add the MDT to the OBX-3-1 in 6.2.1.6.1
2	ST	199	Text	M	<Calculation Description>	▪ Same as the OBX-3-2 of 6.2.1.6.1
3	ID	20	Name of Coding System	M	<Calculation Code System>	▪ Same as the OBX-3-3 of 6.2.1.6.1
4	ST	20	Alternate Identifier	X		Unsupported
5	ST	199	Alternate Text	X		Unsupported
6	ID	20	Name of Alternate Coding System	X		Unsupported

OBX-5 Observation Value

Set the added information for the calculation result

■Component table : VALUES Type

SEQ	DT	LEN	COMPONENT NAME	Opt	Value	Description
-	-	-	String Data	M	<Added information>	-

The following added information is written

<Slide Lot Number> <Slide Valid Term> <Test Result Warning Information> <Emergency ID Use Existence or Non-existence> <Slide Lot Validity or Invalidity Information> <Slide Expiration Information>

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Table OBX-5-1 Added information

	Item	Value	Description
1	Slide lot number	<Slide Lot Number>	[Normal slide item] Slide lot code (6 characters) [Calculated Item] No output
2	Slide valid term	<Slide valid term>	[Normal slide item] Slide valid term YYYY/MM [Calculated Item] No output
3	Test result warning information	<Test result warning information>	▪ Refer to the list OBX-5-2 for the placement of each warning character * The part with fixed length and no warnings should be blank (0x20)
4	Emergency ID use existence or non-existence	<Emergency ID use existence or non-existence>	[USM ON] 0: Not used 1: Used [USM OFF] No output
5	Slide lot validity or invalidity information (CHK QC)	<Validity or Invalidity>	[USM ON] 0: Valid 1: Invalid [USM OFF] No output
6	Slide expiration information (EXP)	<Validity or Invalidity>	[USM ON] 0: Valid 1: Invalid [USM OFF] No output

Table OBX-5-2 Warning character

Position (sending order)	Warning character	Description
1st	H	Testing value exceeds the upper limit of the preset reference interval.
	L	Testing value falls below the lower limit of the preset reference interval.
2nd	@	Testing value is outside of the determination range.
3rd	#	The valid term of the slide has expired.
4th	\$	Temperature control error (thermistor disconnection)
	+	Temperature control error (Incubator temperature has exceeded the upper limit of the range.)
	-	Temperature control error (Incubator temperature has fallen below the lower limit of the range.)
5th	*	Photometer malfunction (white plate fluctuation)
	?	Photometer malfunction (out of the white plate range)
6th	Unused	Always a space (20h)
7th	&	Abnormally high testing value
		*1
8th	Unused	Always a space (20h)
9th	E	Testing value is error. (Calculation error or malfunction of ISE test results *1)
10th	\	Un-spotted slide
		*1
11th	Unused	Always a space (20h)

* "&" becomes "\T", and "¥" becomes "\E" because "&" and "¥" overlap with the MSH-1 field delimiter "\" is a backslash and it may be displayed as "¥" in the Japanese language operating system. In that case, replace "\" with "¥". Be careful that the OBX-5-2 displays 11 types of characters, but it does not always display 11 characters.

OBX-11 Observation Result Status

■ Component table : ID Type (String Data)

SEQ	DT	LEN	COMPONENT NAME	Opt	Value	Description
-	-	199	Coded Value for HL7-Defined Tables	M	<Result Status>	▪ Set "X" (fixed)

6.2.1.7 TCD-Test Code Detail Segment

Table 6.2.1.7-1 Test Code Detail Segment

Sequence	Field Name	Data Type	Repeat	Required	Use by Sender
TCD-1	Universal Service Identifier	CE	1	R	Yes
TCD-2	Auto-Dilution Factor	SN	1	O	Yes
TCD-3	Rerun Dilution Factor	SN	1	O	No
TCD-4	Pre-Dilution Factor	SN	1	O	No
TCD-5	Endogenous Content of Pre-Dilution Diluent	SN	1	O	No
TCD-6	Automatic Repeat Allowed	ID	1	O	No
TCD-7	Reflex Allowed	ID	1	O	No
TCD-8	Analyte Repeat Status	CE	1	O	Yes

TCD-1 Universal Service Identifier

Set the test item number. Set the same value as OBX-3.

■ Component Table: CE Type

SEQ	DT	LEN	COMPONENT NAME	Opt	Value	Description
1	ST	20	Identifier	M	<Calculation Code>	▪ Set the first OBX-3 element
2	ST	199	Text	M	<Calculation Description>	▪ Set the second OBX-3 element
3	ID	20	Name of Coding System	M	<Calculation Code System>	▪ Set the third OBX-3 element
4	ST	20	Alternate Identifier	X		Unsupported
5	ST	199	Alternate Text	X		Unsupported
6	ID	20	Name of Alternate Coding System	X		Unsupported

2

TCD-2 Auto-Dilution Factor

Set the dilution ratio used during measurement.

■ Component Table: SN Type

SEQ	DT	LEN	COMPONENT NAME	Opt	Value	Description
1	ST	2	Comparator	X		▪ Specify as “empty” because it is the value as the default “=”
2	NM	15	Num1	M	1	▪ Set “1” (fixed)
3	ST	1	Separator/Suffix	M	:	▪ Set “.” (fixed)
4	NM	15	Num2	M	<Dilution Ratio During Measurement>	-

TCD-8 Analyte Repeat Status

Set the analyte repeat status according to the measurement status

■ Component Table: CE Type

SEQ	DT	LEN	COMPONENT NAME	Opt	Value	Description
1	ST	20	Identifier	M	<Analyte Repeat Status Code Values>	▪ Set the code value in Table TCD-8-1
2	ST	199	Text	M	<Analyte Repeat Status Code Value Description>	▪ Set description in Table TCD-8-1
3	ID	20	Name of Coding System	M	<Analyte Repeat Status Code Values>	▪ Set the code system in Table TCD-8-1
4	ST	20	Alternate Identifier	X		Unsupported
5	ST	199	Alternate Text	X		Unsupported
6	ID	20	Name of Alternate Coding System	X		Unsupported

Table TCD-8-1 Analyte Repeat Status

Measurement Status	Code Value	Description	Code System
Initial (all except re-test)	O	Original, first run	HL70389
Re-test (with dilution)	D	Repeated with dilution	HL70389
Re-test (without dilution)	R	Repeated without dilution	HL70389

6.2.2 Example Messages

Example of a message when sending test results of the following tests:

Test time : 2020/01/0815:36

No. (Sample No.) : 123456799

Sample ID : ID000001

Specimen Type : Blood Serum (selected in the mode function)

Sex : Male

Operator ID : MYID001

Test item : GLU, TCHO, HDLC

Calculated Item : non-HDL-C

<USM ON>

```
MSH|^~&|123456799|20200108155010|OUL^R22^OUL_R22|1|P|2.5.1|||||UNICODE UTF-8||LAB-29^IHE
SPM|1|ID000001|SER^Serum^HL70487|||||P^Patient^HL70369
SAC|||123456799|||||1^A^L
OBR|||U^Unspecified identifier^L
ORC|SC|||CM
OBX|1|NM|3D010000002329101^GLU^JC10||105|mg/dL^UCUM|73-109|F||||MYID001|SN0123456789|20200108153601
OBX|1|ST|3D010000002329101^TMDT^GLU^JC10||123456,2018/12, # ,1,0,0||||F
TCD|3D010000002329101^GLU^JC10|^1^1||||O^Original, first run^HL70389
OBX|2|NM|3F050000002329101^TCHO^JC10||150|mg/dL^UCUM|F||||MYID001|SN0123456789|20200108153601
OBX|2|ST|3F050000002329101^TMDT^TCHO^JC10||223344,2020/12, ,1,0,0||||F
TCD|3F050000002329101^TCHO^JC10|^1^1||||O^Original, first run^HL70389
OBX|3|NM|3F070000002329101^HDL-C^JC10||50|mg/dL^UCUM|F||||MYID001|SN0123456789|20200108153601
OBX|3|ST|3F070000002329101^TMDT^HDL-C^JC10||234567,2020/12, ,1,0,0||||F
TCD|3F070000002329101^HDL-C^JC10|^1^1||||O^Original, first run^HL70389
OBX|4|NM|3F069000002391901^non-HDL-C^JC10||100|mg/dL^UCUM|F||||MYID001|SN0123456789|20200108153601
OBX|4|ST|3F069000002391901^TMDT^non-HDL-C^JC10||, ,1,0,0||||F
TCD|3F069000002391901^non-HDL-C^JC10|^1^1||||O^Original, first run^HL70389
```

<USM OFF>

```
MSH|^~&|123456799|20200108155010|OUL^R22^OUL_R22|1|P|2.5.1|||||UNICODE UTF-8||LAB-29^IHE
SPM|1|ID000001|SER^Serum^HL70487|||||P^Patient^HL70369
SAC|||123456799|||||1^A^L
OBR|||U^Unspecified identifier^L
ORC|SC|||CM
OBX|1|NM|3D010000002329101^GLU^JC10||105|mg/dL^UCUM|73-109|F||||MYID001|SN0123456789|20200108153601
OBX|1|ST|3D010000002329101^TMDT^GLU^JC10||123456,2018/12, # ,,,||||F
TCD|3D010000002329101^GLU^JC10|^1^1||||O^Original, first run^HL70389
OBX|2|NM|3F050000002329101^TCHO^JC10||150|mg/dL^UCUM|F||||MYID001|SN0123456789|20200108153601
OBX|2|ST|3F050000002329101^TMDT^TCHO^JC10||223344,2020/12, ,,,||||F
TCD|3F050000002329101^TCHO^JC10|^1^1||||O^Original, first run^HL70389
OBX|3|NM|3F070000002329101^HDL-C^JC10||50|mg/dL^UCUM|F||||MYID001|SN0123456789|20200108153601
OBX|3|ST|3F070000002329101^TMDT^HDL-C^JC10||234567,2020/12, ,,,||||F
TCD|3F070000002329101^HDL-C^JC10|^1^1||||O^Original, first run^HL70389
OBX|4|NM|3F069000002391901^non-HDL-C^JC10||100|mg/dL^UCUM|F||||MYID001|SN0123456789|20200108153601
OBX|4|ST|3F069000002391901^TMDT^non-HDL-C^JC10||, ,,,||||F
TCD|3F069000002391901^non-HDL-C^JC10|^1^1||||O^Original, first run^HL70389
```

6.3 ACK-General Acknowledgment message

- Messages between equipment shall be sent in both directions. However, only received in the current version.

The message sent by the Sender is received by the Receiver, which returns an acceptance result with this message.

- Message transmission/operation flow shall reference “5.2.1 Send Message”

Table 6.3-1 ACK Message

Segment		Required	Use by Sender Or Receiver
MSH	Message Header segment	R	Yes
[[SFT]]	Software segment	O	No
MSA	Message Acknowledgement segment	R	Yes
[[ERR]]	Error segment	O	Yes(if exists)

6.3.1 Used Segment

6.3.1.1 MSH-Message Header Segment

Data organization references “6.2.1.1 MSH-Message Header Segment”.

The content shall be as follows for the MSH of an Ack message.

Sequence	FDC Processing Content/Check Content
MSH-1	Interpreted based on HI7 specifications
MSH-2	Same as above
MSH-5	Shall be the same value as the MSH-3 Sending Application of the message sent by FDC
MSH-6	Shall be the same value as the MSH-4 Sending Facility of the message sent by FDC
MSH-7	Read in the format of *1
MSH-11	Shall be the same value as the MSH-11 Processing ID of the message sent by FDC
MSH-12	No version restrictions
MSH-21	Interpreted based on HI7 specifications

*1: YYYY[MM[DD[HH[MM[SS[.S[S[S[S]]]]]]]]][+/-ZZZZ]

6.3.1.2 MSA-Message Acknowledgment Segment

Table 6.3.1.2-1 Message Acknowledgment Segment

Sequence	Field Name	Data Type	Repeat	Required	Use by Sender or Receiver
MSA-1	Acknowledgment Code	ID	1	R	Yes
MSA-2	Message Control ID	ST	1	R	Yes
MSA-3	Text Message	ST	1	B	No
MSA-4	Expected Sequence Number	NM	1	O	
MSA-5	Delayed Acknowledgment Type	-	1	W	No
MSA-6	Error Condition	CE	1	B	No

MSA-1 Acknowledgment Code

Set Ack code using HL7 table0008.

■ Component Table: ID Type (String Data)

SEQ	DT	LEN	COMPONENT NAME	Opt	Value	Description
-	-	199	Coded Value for HL7-Defined Tables	M	<ACK Code Value>	▪ Set the value in Table MSA-1-1

Table MSA-1-1 Acknowledgment Code

Value	Description	Sender	Receiver
AA	Original mode: Application Accept - Enhanced mode: Application acknowledgment: Accept	Set based on HL7 specifications	Operate based on "5.2.1.2 Receive Ack (for receive Ack)"
AE	Original mode: Application Error - Enhanced mode: Application acknowledgment: Error		
AR	Original mode: Application Reject - Enhanced mode: Application acknowledgment: Reject		
CA	Enhanced mode: Accept acknowledgment: Commit Accept		
CE	Enhanced mode: Accept acknowledgment: Commit Error		
CR	Enhanced mode: Accept acknowledgment: Commit Reject		

MSA-2 Message Control ID

■ Component table: ST type (String Data)

SEQ	DT	LEN	COMPONENT NAME	Opt	Value	Description
-	-	199	String Data	M	<Message Control ID>	Check whether it is the same value as the message control ID (MSH-10) of the sent message because determination of Ack for the sent message occurs when ACK is sent

6.3.1.3 ERR-Error Segment

Table 6.3.1.3-1 Error Segment

Sequence	Field Name	Data Type	Repeat	Required	Use by Sender or Receiver
ERR-1	Error Code and Location	ELD	*	B	No
ERR-2	Error Location	ERL	*	O	No
ERR-3	HL7 Error Code	CWE	1	R	Yes
ERR-4	Severity	ID	1	R	Yes
ERR-5	Application Error Code	CWE	1	O	No
ERR-6	Application Error Parameter	ST	*	O	No
ERR-7	Diagnostic Information	TX	1	O	No
ERR-8	User Message	TX	1	O	No
ERR-9	Inform Person Indicator	IS	*	O	No
ERR-10	Override Type	CWE	1	O	No
ERR-11	Override Reason Code	CWE	*	O	No
ERR-12	Help Desk Contact Point	XTN	*	O	No

ERR-3 HL7 Error Code

Set error code using HL7 table0357.

■ Component Table: CWE Type

SEQ	DT	LEN	COMPONENT NAME	Opt	Value	Description
1	ST	20	Identifier	M	<Error Code Value>	▪ Set the value in Table ERR-3-1
2	ST	199	Text	M	<Error Code Content>	▪ Set description in Table ERR-3-1
3	ID	20	Name of Coding System	M	<Error Code System>	▪ Set "HL70357" (fixed)
4	ST	20	Alternate Identifier	X		Unsupported
5	ST	199	Alternate Text	X		Unsupported
6	ID	20	Name of Alternate Coding System	X		Unsupported
7	ST	10	Coding System Version ID	X		Unsupported
8	ST	10	Alternate Coding System Version ID	X		Unsupported
9	ST	199	Original Text	X		Unsupported

Table ERR-3-1 Error Code - Identifier

Value	Description	Sender	Receiver
0	Message accepted	Set based on HL7 specifications	Save information in the log
100	Segment sequence error		
101	Required field missing		
102	Data type error		
103	Table value not found		
200	Unsupported message type		
201	Unsupported event code		
202	Unsupported processing id		
203	Unsupported version id		
204	Unknown key identifier		
205	Duplicate key identifier		
206	Application record locked		
207	Application internal error		

ERR-4 Severity

Supports HL7 table0516 error severity

■ Componen Table: ID Type (String Data)

SEQ	DT	LEN	COMPONENT NAME	Opt	Value	Description
-	-	199	Coded Value for HL7-Defined Tables	M	<Error Type>	▪ Set the value in Table ERR-4-1

Table ERR-4-1 Severity

Value	Description	Sender	Receiver
W	Warning	Set based on HL7 specifications	Save information in the log
I	Information		
E	Error		

End

7. Appendix 2 FDC HL7 sample type code (LOINC)

				LOINC Code (is chosen based on "Unit Conversion" and "Slide Code")										
				NX600 Unit Conversion		PS			W		U			
NX600 Slide Code				Unit (A)	Unit (B)		Unit (A)	Unit (B)		Unit (A)	Unit (A)	Unit (A)	Unit (B)	
Test Name	Test code	Sample code												
Single Item	GLU	10	50/0	mg/dL	mmol/L		2345-7	14749-6		2339-0	15074-8	2350-7	15076-3	
	BUN	11	50	mg/dL	mmol/L		3094-0	14937-7		-	-	3095-7	14938-5	
	UREA	11	50	mg/dL	mmol/L		3091-6	22664-7		-	-	3092-4	22700-9	
	Hb	12	0	g/dL	g/L		-	-		718-7	-	-	-	
	UA	13	50	mg/dL	μmol/L		3084-1	14933-6		-	-	3086-6	14934-4	
	TCHO	14	50	mg/dL	mmol/L		2093-3	14647-2		-	-	14444-4	(No definition)*	
	NH3	15	50/0	μmg/dL	μmol/L		22763-7	16362-6		32664-5	1839-0	20682-1	1842-4	
	TG	16	50	mg/dL	mmol/L		2571-8	14927-8		-	-	14450-1	59572-8	
	CRE	17	50	mg/dL	μmol/L		2160-0	14682-9		-	-	2161-8	14683-7	
	TP	18	50	g/dL	g/L		2885-2		-		-	-	(No definition)*	
	ALB	20	50	g/dL	g/L	μmol/L	1751-7	1751-7	54347-0	-	-	1754-1	1754-1	(No definition)*
	TBIL	21	50	mg/dL	μmol/L		1975-2	14631-6		-	-	1978-6	70199-5	
	Ca	23	50	mg/dL	mmol/L	mEq/L	17861-6	2000-8		-	-	17862-4	2004-0	
	IP	24	50	mg/dL	mmol/L		2777-1	14879-1		-	-	2778-9	13539-2	
	DBIL	25	50	mg/dL	μmol/L		1968-7	14629-0		-	-	-	(No definition)*	
	HDL-C	26	50	mg/dL	mmol/L		2085-9	14646-4		-	-	-	(No definition)*	
	Mg	28	50	mg/dL	mmol/L		19123-9	2601-3		-	-	19124-7	2598-1	
	GGT	30	50	U/L	μkat/L	U/L	2324-2		-		-	-	27403-5	
	GOT/AST	31	50	U/L	μkat/L	K-U	1920-8		-		-	-	14413-9	
	GPT/ALT	32	50	U/L	μkat/L	K-U	1742-6		-		-	-	(No definition)*	
	CPK	33	50	U/L	μkat/L	U/L	2157-6		-		-	-	(No definition)*	
	LDH	34	50	U/L	μkat/L	WL-U	14804-9		-		-	-	2534-6	
	ALP	35	50	U/L	μkat/L	KA-U	6768-6		-		-	-	16337-8	
	LAP	37	50	U/L	μkat/L	U/L	2187-3		-		-	-	2189-9	
	CKMB	38	50	U/L	μkat/L	U/L	32673-6		-		-	-	(No definition)*	
	CHE	39	50	U/L	μkat/L	U/L	2098-2		-		-	-	(No definition)*	
	LDHi	40	50	U/L	μkat/L	WL-U	14804-9		-		-	-	2534-6	
	ALPi	41	50	U/L	μkat/L	KA-U	6768-6		-		-	-	16337-8	
	AMYL	43	50	U/L	μkat/L	S-U	1798-8		-		-	-	1799-6	
	v-AMY	43	50	U/L	μkat/L		1798-8		-		-	-	1799-6	
	LIP	44	50	U/L	μkat/L		2572-6		-		-	-	(No definition)*	
	TCO2	45	50	mmol/L			2028-9		-		-	-	13538-4	
	CRP	51	50	mg/dL	mg/L		1988-5		-		-	-	(No definition)*	
	vc-CRP	52	50	mg/dL	mg/L		1988-5		-		-	-	(No definition)*	
	Na	91	-	mEq/L	mmol/L		2951-2		2947-0		-	-	2955-3	
	K	92	-	mEq/L	mmol/L		2823-3		6298-4		-	-	2828-2	
	Cl	93	-	mEq/L	mmol/L		2075-0		2069-3		-	-	2078-4	
Calculated Item	LDL-C	101	-	mg/dl	mmol/l		13457-7	39469-2		-	-	(No definition)*		
	non-HDL-C	102	-	mg/dl	mmol/l		43396-1	70204-3		-	-	(No definition)*		
	GLOB	103	-	g/dl	g/l		10834-0		-		-	18320-2		
	ALB/GLOB	104	-	-		1759-0		-		-	-	9405-2		
	BUN/CRE	105	-	mg/mg	(SI)		3097-3	(No definition)*		-	-	44416-6	(No definition)*	
	UREA / CRE	105	-	mg/mg	(SI)		56997-0	39783-6		-	-	(No definition)*	59786-4	
	GOT/GPT(AST/ALT)	106	-	-		1916-6		-		-	-	(No definition)*		
	Na/K	107	-	-		28003-2		-		-	-	2959-5		
	Anion Gap	108	-	mEq/l	mmol/l		33037-3		-		-	-	(No definition)*	

*For parameters of "No definition", "Parameter name^Parameter name^L" is outputted.

8. Appendix 3 Relationship between Measurement Results and OBX-2, 5, 8, 11

Pattern	Out of range	Warning characters	Lot validity or invalidity	Measurement result (reprint data)				HL7						
				Measurement value result_char	Sign result_sign	Warning characters Err	ISE error ise_err	OBX-2 Value Type	OBX-5 Observation Value	OBX-8 Abnormal Flags	OBX-11 Observation Result Status	OBX-5-1 Test result warning information	OBX-5-1 Slide lot validity or invalidity information 0: Valid 1: Invalid	OBX-5-1 Slide expiration information 0: Valid 1: Invalid
Normal	None	None	Valid	Measurement value	=	None	None	NM	Measurement value	""	F	None	0	0
			Invalid	CHK QC	=	None	None	NM	""	A	X	None	1	0
		*. #(with measurement value) *1	Valid	Measurement value	=	*. #	None	NM	Measurement value	A	F	*. #	0	0
			Invalid	CHK QC	=	*. #	None	NM	""	A	X	*. #	1	0
		EXP	Valid	EXP	=	None	None	NM	""	A	X	None	0	1
Out of reference interval range	Measurement value is higher than the upper limit of the preset reference interval.		Invalid	EXP	=	None	None	NM	""	A	X	None	1	1
		None	Valid	Measurement value	=	H	None	NM	Measurement value	H	F	H	0	0
			Invalid	CHK QC	=	H	None	NM	""	H~A	X	H	1	0
		*. #(with measurement value) *1	Valid	Measurement value	=	H, *. #	None	NM	Measurement value	H~A	F	H, *. #	0	0
			Invalid	CHK QC	=	H, *. #	None	NM	""	H~A	X	H, *. #	1	0
		EXP	Valid	EXP	=	H	None	NM	""	H~A	X	H	0	1
	Measurement value is lower than the lower limit of the preset reference interval.		Invalid	EXP	=	H	None	NM	""	H~A	X	H	1	1
		None	Valid	Measurement value	=	L	None	NM	Measurement value	L	F	L	0	0
			Invalid	CHK QC	=	L	None	NM	""	L~A	X	L	1	0
		*. #(with measurement value) *1	Valid	Measurement value	=	L, *. #	None	NM	Measurement value	L~A	F	L, *. #	0	0
			Invalid	CHK QC	=	L, *. #	None	NM	""	L~A	X	L, *. #	1	0
		EXP	Valid	EXP	=	L	None	NM	""	L~A	X	L	0	1
Out of measurement range	Measurement value is higher than indicative value. Measurement value is higher than the upper limit of the measurement range. (<>Display setting)		Invalid	EXP	=	L	None	NM	""	L~A	X	L	1	1
		None	Valid	Upper limit of measurement range	>	None	None	NM	""	>	X	None	0	0
			Invalid	CHK QC	>	None	None	NM	""	>~A	X	None	1	0
		*. #(with measurement value) *1	Valid	Upper limit of measurement range	>	*. #	None	NM	""	>~A	X	*. #	0	0
			Invalid	CHK QC	>	*. #	None	NM	""	>~A	X	*. #	1	0
		EXP	Valid	EXP	>	None	None	NM	""	>~A	X	None	0	1
	Measurement value is higher than the upper limit of the measurement range. (@Display setting)		Invalid	EXP	>	None	None	NM	""	>~A	X	None	1	1
		None	Valid	Measurement value	=	@	None	NM	Measurement value	>	F	@	0	0
			Invalid	CHK QC	=	@	None	NM	""	>~A	X	@	1	0
		*. #(with measurement value) *1	Valid	Measurement value	=	@, *. #	None	NM	Measurement value	>~A	F	@, *. #	0	0
			Invalid	CHK QC	=	@, *. #	None	NM	""	>~A	X	@, *. #	1	0
		EXP	Valid	EXP	=	@	None	NM	""	>~A	X	@	0	1
	Measurement value is lower than the lower limit of the measurement range. (@Display setting)		Invalid	EXP	=	@	None	NM	""	>~A	X	@	1	1
		None	Valid	Measurement value	=	@	None	NM	Measurement value	<	F	@	0	0
			Invalid	CHK QC	=	@	None	NM	""	>~A	X	@	1	0
		*. #(with measurement value) *1	Valid	Measurement value	=	@, *. #	None	NM	Measurement value	<~A	F	@, *. #	0	0
			Invalid	CHK QC	=	@, *. #	None	NM	""	<~A	X	@, *. #	1	0
		EXP	Valid	EXP	=	@	None	NM	""	<~A	X	@	0	1
Out of measurement range in ISE	Measurement value is higher than the upper limit of the measurement range.		Invalid	EXP	=	@	None	NM	""	<~A	X	@	1	1
		None	Valid	Lower limit of measurement range	<	None	None	NM	""	<	X	None	0	0
			Invalid	CHK QC	<	None	None	NM	""	<~A	X	None	1	0
		*. #(with measurement value) *1	Valid	Lower limit of measurement range	<	*. #	None	NM	""	<~A	X	*. #	0	0
			Invalid	CHK QC	<	*. #	None	NM	""	<~A	X	*. #	1	0
		EXP	Valid	EXP	<	None	None	NM	""	<~A	X	None	0	1
	Measurement value is lower than the lower limit of the measurement range. (<>Display setting) Measurement value is lower than the display digit minimum value.		Invalid	EXP	<	None	None	NM	""	<~A	X	None	1	1
		None	Valid	+OR	>	None	3	NM	""	>	X	E	0	0
			Invalid	CHK QC	>	None	3	NM	""	>~A	X	E	1	0
		EXP	Valid	EXP	>	None	3	NM	""	>~A	X	E	0	1
			Invalid	EXP	>	None	3	NM	""	>~A	X	E	1	1
		Measurement value is lower than the lower limit of the measurement range.	Valid	-OR	<	None	3	NM	""	<	X	E	0	0
No measurement value	None		Invalid	CHK QC	<	None	3	NM	""	<~A	X	E	1	0
		None	Valid	Measurement value	=	\$, \, ?, E, &, *, +, - (No Measurement value)	None	NM	""	A	X	\$, \, ?, E, &, *, +, -	0	0
			Invalid	CHK QC	=	\$, \, ?, E, &, *, +, -	None	NM	""	A	X	\$, \, ?, E, &, *, +, -	1	0
		*. #(with measurement value) and *. # (with measurement value)	Valid	Measurement value	=	\$, \, ?, E, &, *, +, -	None	NM	""	A	X	\$, \, ?, E, &, *, +, -	0	0
			Invalid	CHK QC	=	\$, \, ?, E, &, *, +, -	None	NM	""	A	X	\$, \, ?, E, &, *, +, -	1	0
	None	\$, \, ?, E, &, *, +, - (No Measurement value) and *. # (with measurement value) and EXP	Valid	EXP	=	\$, \, ?, E, &, *, +, -	None	NM	""	A	X	\$, \, ?, E, &, *, +, -	0	1
			Invalid	EXP	=	\$, \, ?, E, &, *, +, -	None	NM	""	A	X	\$, \, ?, E, &, *, +, -	1	1
		None	Valid	Measurement value	=	None	None	NM	""	""	X	None	0	0
			Invalid	CHK QC	=	None	None	NM	""	A	X	None	1	0
		E1,E2,E4,E5,E6,E7,E8	Valid	Measurement value	=	None	1,2,4,5,6,7,8	NM	""	A	X	E	0	0
ISE error	None		Invalid	CHK QC	=	None	1,2,4,5,6,7,8	NM	""	A	X	E	1	0
		E1,E2,E4,E5,E6,E7,E8	Valid	Measurement value	=	None	1,2,4,5,6,7,8	NM	""	A	X	E	0	0
			Invalid	CHK QC	=	None	1,2,4,5,6,7,8	NM	""	A	X	E	1	0
		E1,E2,E4,E5,E6,E7,E8 and +, -, \$ (No Measurement value)	Valid	Measurement value	=	+, -, \$	1,2,4,5,6,7,8	NM	""	A	X	+, -, \$, E	0	0
	E1,E2,E4,E5,E6,E7,E8 and # (with measurement value)		Invalid	CHK QC	=	+, -, \$	1,2,4,5,6,7,8	NM	""	A	X	+, -, \$, E	1	0
		None	Valid	Measurement value	=	#	1,2,4,5,6,7,8	NM	""	A	X	#, E	0	0
			Invalid	CHK QC	=	#	1,2,4,5,6,7,8	NM	""	A	X	#, E	1	0
		E1,E2,E4,E5,E6,E7,E8 and EXP	Valid	EXP	=	None	1,2,4,5,6,7,8	NM	""	A	X	E	0	1
	None		Invalid	EXP	=	None	1,2,4,5,6,7,8	NM	""	A	X	E	1	1

*1 Including coincidence.